MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW POWER SCHOTTKY TTL, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low power Schottky TTL, bus transceivers with three-state outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. Device type shall be as follows:

Device type	Circuit
01	Quadruple inverting bus transceivers with three-state outputs
02	Quadruple noninverting bus transceivers with three-state outputs
03	Octal noninverting bus transceivers with three-state outputs
04	Octal inverting bus transceivers and registers with three-state outputs
05	Octal inverting bus transceivers and registers with three-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat-package
R	D-8 (20-lead, 1/4 x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat-package
L	D-9 (24-lead, 1/4" x 1 1/4"), dual-in-line package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01 and 02	297 mW dc
Device type 03	522.5 mW dc
Device types 04 and 05	907.5 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases A and D	70°C/W
Case C	50°C/W
Cases R and L	40°C/W
Cases S and 2	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	
Device types 01, 02, and 03	0.7 V dc
Device types 04 and 05	0.5 V dc
Normalized fanout (each input) 2/	20 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Width of clock pulse (t_{CLK})	
Device types 04 and 05	20 ns
Setup time before clock (t_{setup})	
Device types 04 and 05	20 ns
Hold time (t_{hold})	
Device types 04 and 05	0 ns

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connection. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be in accordance with 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

2/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that beihg tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH1}	V_{CC} = 4.5 V, I_{OH} = -3 mA, V_{IH} = 2.0 V, V_{IL} = 0.7 V	01,02 03	2.4	---	V
		V_{IL} = 0.5 V	04, 05	2.4	---	"
	V_{OH2}	V_{CC} = 4.5 V, I_{OH} = -12 mA, V_{IH} = 2.0 V, V_{IL} = 0.5 V	01,02 03	2.0	---	"
		V_{IL} = 0.5 V	04,05	2.0	---	"
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 12 mA, V_{IH} = 2.0 V, V_{IL} = 0.7 V	01,02 03	---	0.4	"
		V_{IL} = 0.5 V	04,05	---	0.4	"
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = 25°C	All	---	-1.5	"
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All	---	20	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All	---	0.1	mA
Inhibited-state output leakage current	I_{OZH}	V_{CC} = 5.5 V, V_{OUT} = 2.7 V	01,02	---	40	µA
			03	---	20	µA
			04,05	---	20	µA
	I_{OZL}	V_{CC} = 5.5 V, V_{OUT} = 0.4 V	01,02 03	---	-200	µA
			04,05	---	-400	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01,02 03	0	-200	µA
			04,05	0	-400	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	All	-40	-225	mA
Supply current	I_{CCH}	V_{CC} = 5.5 V	01,02	---	38	mA
			03	---	70	
			04,05	---	145	
	I_{CCL}	V_{CC} = 5.5 V	01,02	---	50	mA
			03	---	90	
			04,05	---	165	
	I_{CCZ}	V_{CC} = 5.5 V	01,02		50	mA
			03		95	
			04,05		165	
Propagation delay time, low-to-high clock to bus	t_{PLH1}	V_{CC} = 5.0 V, R_L = 110Ω, C_L = 50 pF	04,05	2	39	ns
Propagation delay time, high-to-low clock to bus	t_{PHL1}		04	2	52	ns
			05	2	59	ns
Propagation delay time, low-to-high bus to bus	t_{PLH2}		01,02	2	25	ns
			03	2	22	
			04,05	2	30	
Propagation delay time, high-to-low bus to bus	t_{PHL2}		01,02	2	30	ns
			03	2	22	
			04	2	33	
			05	2	39	

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, low-to-high select (with bus input high) to bus	t_{PLH3}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110Ω	04	2	59	ns
			05	2	78	
Propagation delay time, high-to-low select (with bus input high) to bus	t_{PHL3}		04	2	52	ns
			05	2	59	
Propagation delay time, low-to-high select (with bus input low) to bus	t_{PLH4}		04	2	72	ns
			05	2	59	
Propagation delay time, high-to-low select (with bus input low) to bus	t_{PHL4}		04	2	39	ns
			05	2	59	
Propagation delay time, disabled-to-high level output	t_{PZH1}		01,02	2	36	ns
			03	2	58	
Propagation delay time, disabled-to-high level output enable to bus	t_{PZH2}		04	2	78	ns
			05	2	72	
Propagation delay time, disabled-to-high level output direction to bus	t_{PZH3}		04	2	65	ns
			05	2	59	
Propagation delay time, disabled-to-low level output	t_{PZL1}		01,02	2	46	ns
			03	2	58	
Propagation delay time, disabled-to-low level output enable to bus	t_{PZL2}		04	2	91	ns
			05	2	78	
Propagation delay time, disabled-to-low level direction to bus	t_{PZL3}		04	2	85	ns
			05	2	65	
Propagation delay time, high-level to disabled output	t_{PHZ1}		01,02	2	46	ns
			03	2	39	
Propagation delay time, high-level-to-disabled output enable to bus	t_{PHZ2}		04	2	52	ns
			05	2	65	
Propagation delay time, high-level-to-disabled output direction to bus	t_{PHZ3}		04	2	46	ns
			05	2	52	
Propagation delay time, low-level-to-disabled output	t_{PLZ1}		01,02	2	39	ns
			03	2	39	
Propagation delay time, low-level-to-disabled output enable to bus	t_{PLZ2}		04	2	52	ns
			05	2	52	
Propagation delay time, low-level-to-disabled output direction to bus	t_{PLZ3}		04	2	46	ns
			05	2	46	

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,8, 9,10,11	1*,2,3,7,8,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,7,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).
   (1) Test condition D, using the circuit shown on figure 3, or equivalent.
   (2) T_A = +125°C minimum.
b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

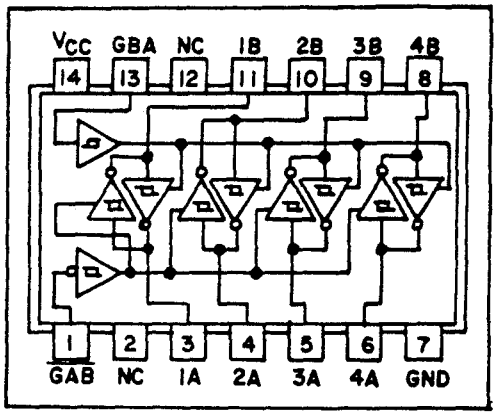
4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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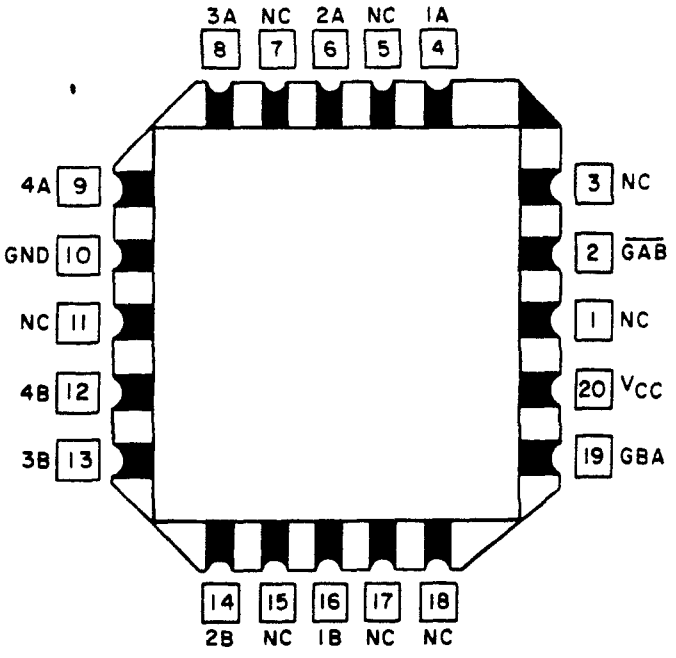
Device type 01

Cases A, C and D



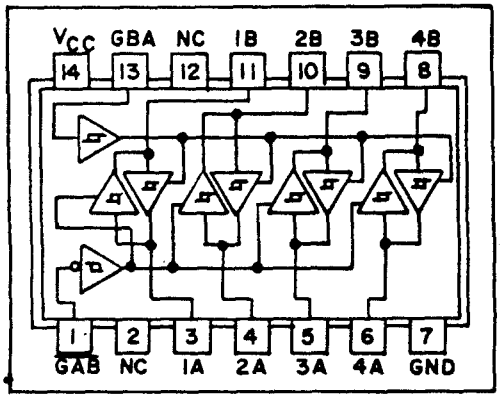
Device type 01

Case 2



Device type 02

Cases A, C and D



Device type 02

Case 2

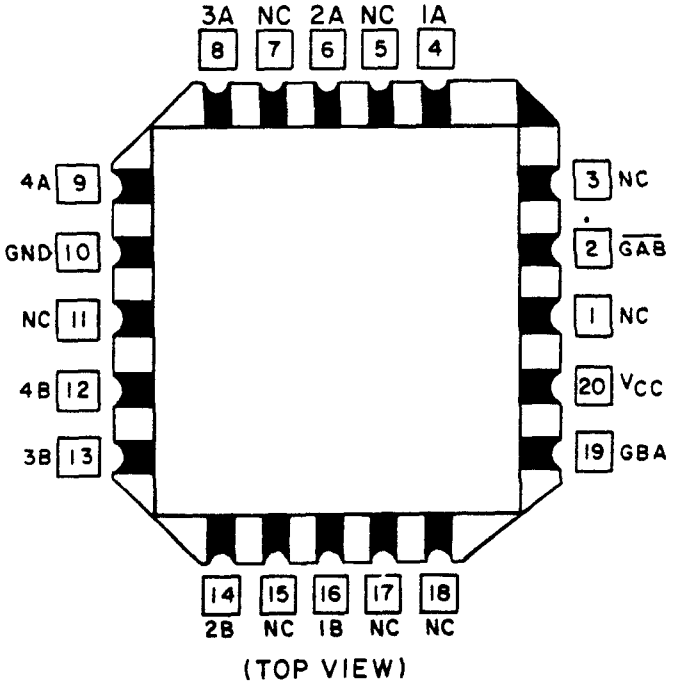
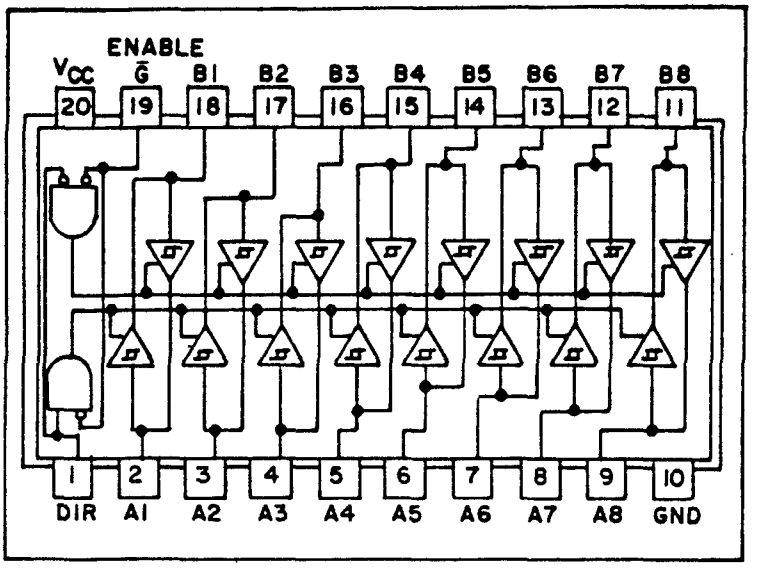


FIGURE 1. Logic diagrams and terminal connections (top view).

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Device type 03

Cases R and S



Device type 03

Case 2

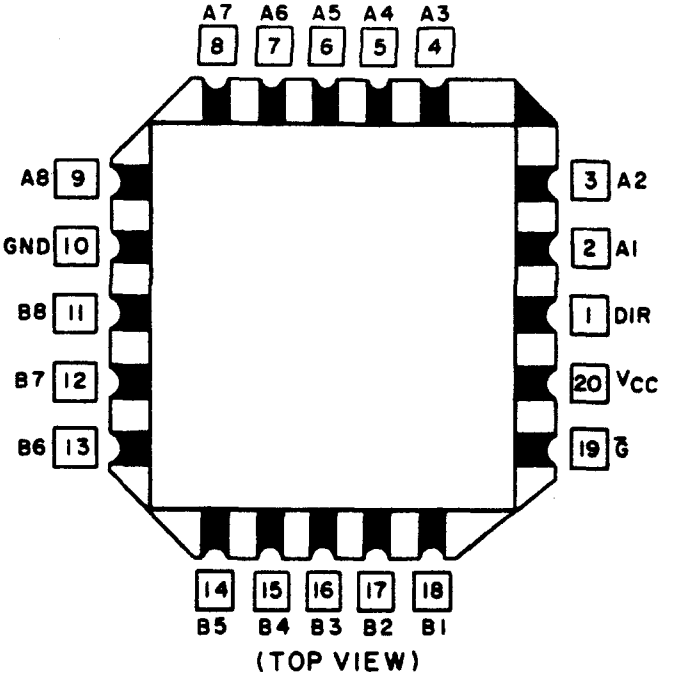


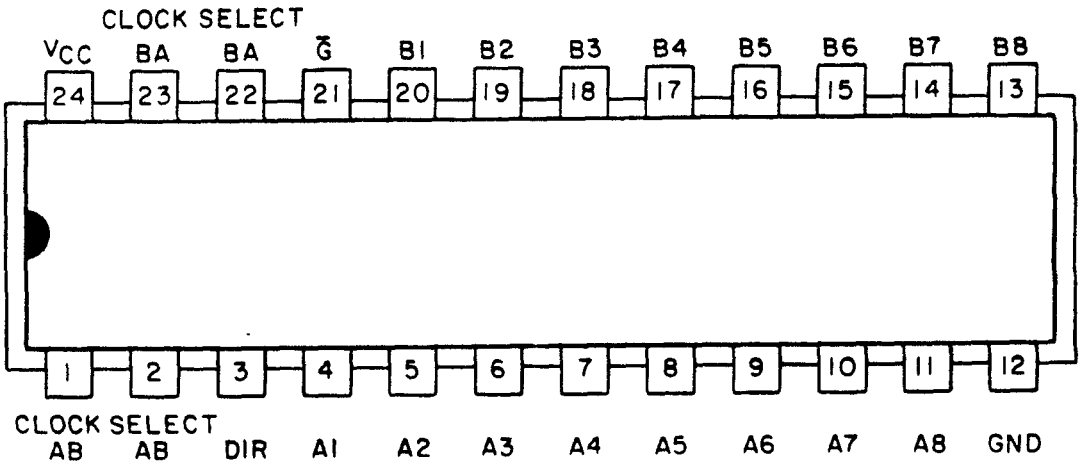
FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device types 04 and 05

(Terminal connections)

Case L



Device type 04

(logic diagram)

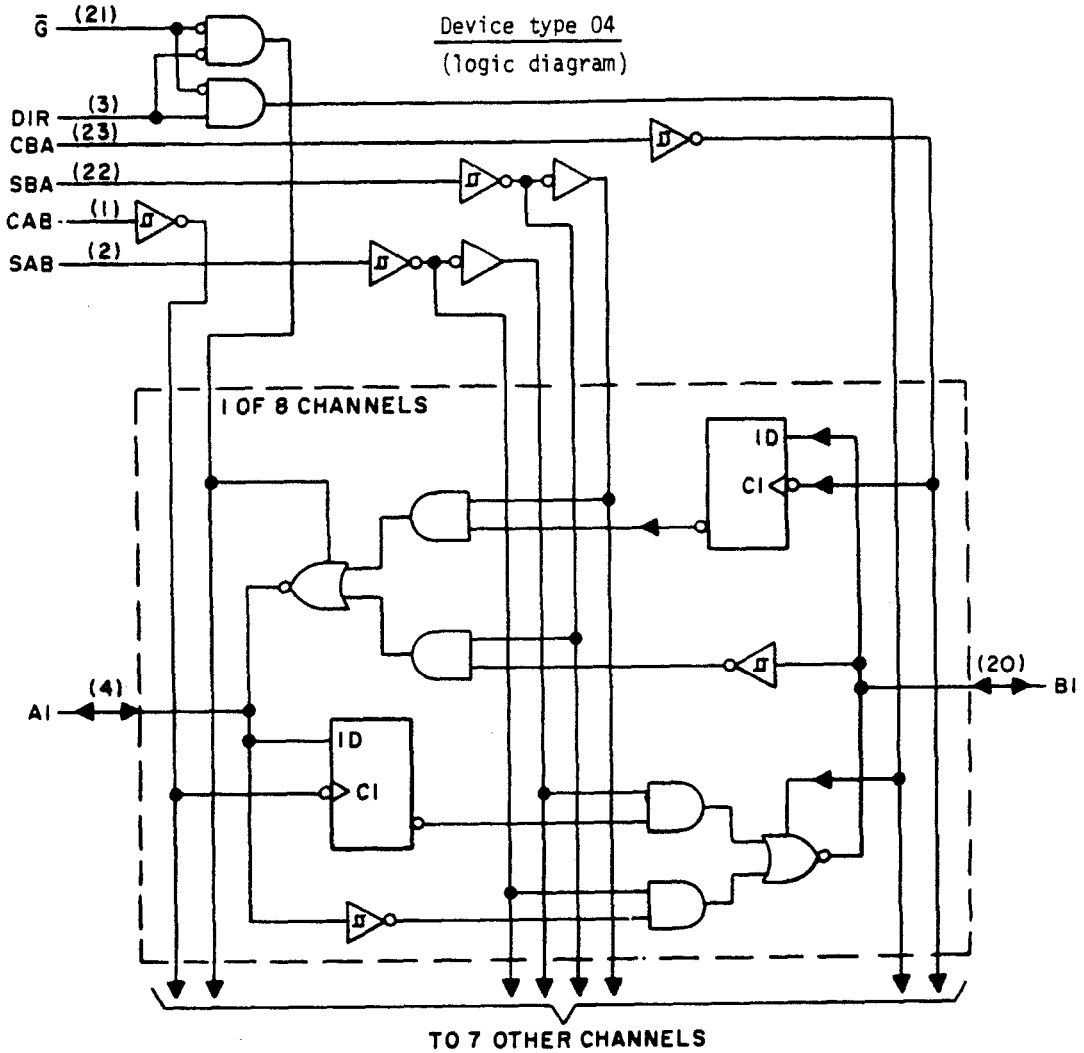


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device type 05
(logic diagrams)

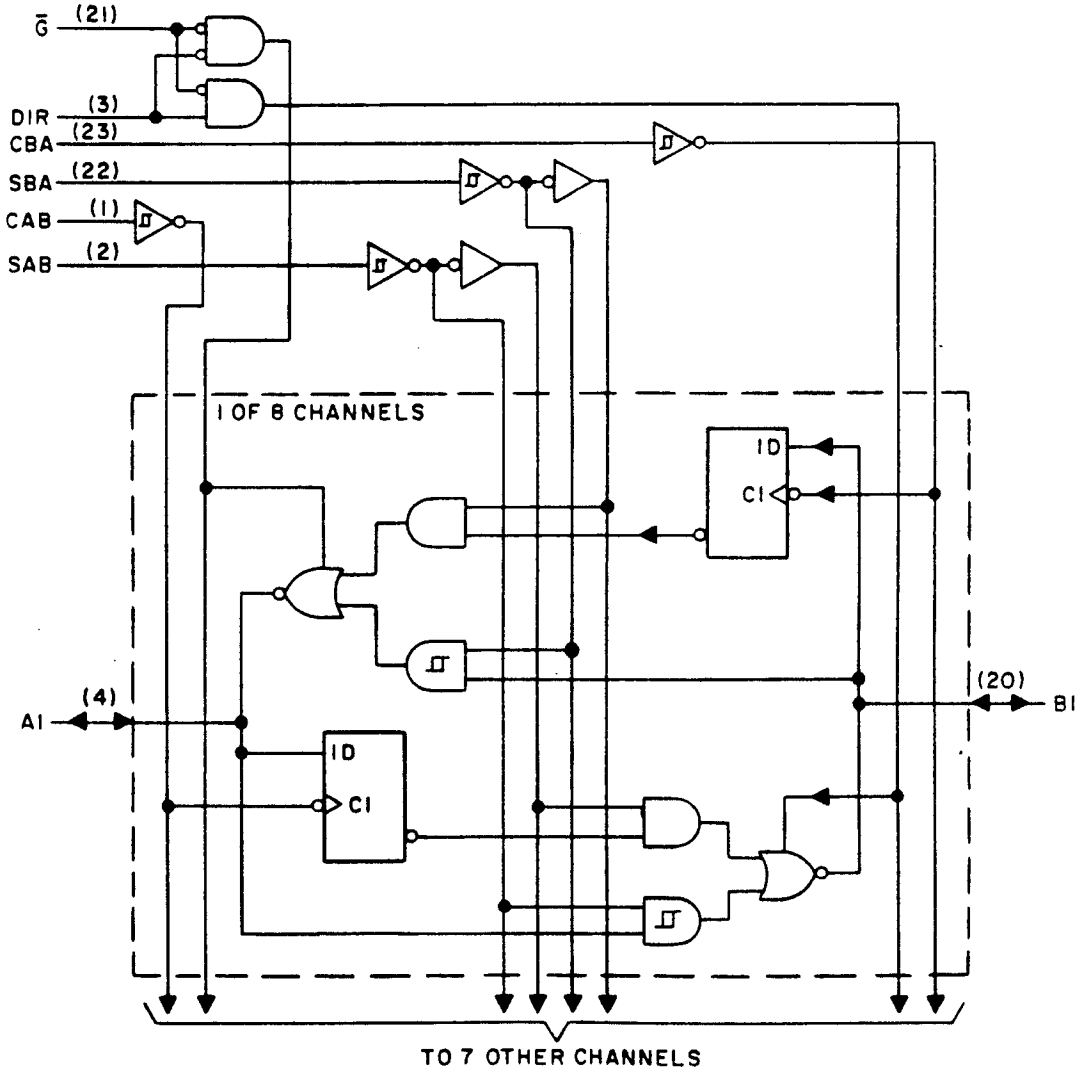


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device type 01

CONTROL INPUTS		DATA PORT STATUS	
$\overline{G}AB$	GBA	A	B
H	H	$\overline{O}$	I
L	H	*	*
H	L	ISOLATED	
L	L	I	$\overline{O}$

*Possibly destructive oscillation may occur if the transceivers are enabled in both directions at once.
I = Input, O = Output, $\overline{O}$ = Inverting Output

Device type 02

CONTROL INPUTS		DATA PORT STATUS	
$\overline{G}AB$	GBA	A	B
H	H	O	I
L	H	*	*
H	L	ISOLATED	
L	L	I	O

*Possibly destructive oscillation may occur if the transceivers are enabled in both directions at once.
I = Input, O = Output, $\overline{O}$ = Inverting Output

Device type 03

ENABLE $\overline{G}$	DIRECTION CONTROL DIR	OPERATION
L	L	B data to A bus
L	H	A data to B bus
H	X	Isolation

H = high level, L = low level, X = irrelevant

FIGURE 2. Truth tables.

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Device type 04

INPUTS						DATA I/O *		OPERATION OR FUNCTION
$\overline{G}$	DIR	CAB	CBA	SAB	SBA	A1 thru A8	B1 thru B8	
H H	X X	H or L ↑	H or L ↑	X X	X X	Input	Input	Isolation Store A and B Data
L L	L L	X X	X X	X X	L H	Output	Input	Real Time B Data to A Bus Stored B Data to A Bus
L L	H H	X H or L	X X	L H	X X	Input	Output	Real Time A Data to B Bus Stored A Data to B Bus

H = High Level L = Low Level X = Irrelevant ↑ = Low-to-high level transition
*The data output functions may be enabled or disabled by various signals at the $\overline{G}$ and DIR inputs. Data input functions are always enabled, i.e., data at the bus pins will be stored on every low-to-high transition on the clock inputs.

Device type 05

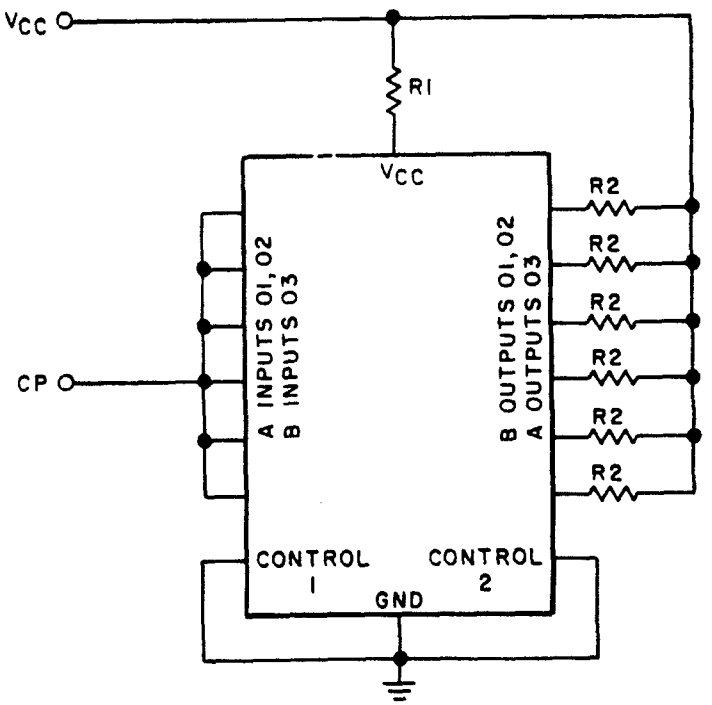
INPUTS						DATA I/O*		OPERATION OR FUNCTION
$\overline{G}$	DIR	CAB	CBA	SAB	SBA	A1 thru A8	B1 thru B8	
H H	X X	H or L ↑	H or L ↑	X X	X X	Input	Input	Isolation Stor A and B Data
L L	L L	X X	X X	X X	L H	Output	Input	Real Time $\overline{B}$ Data to A Bus Stored $\overline{B}$ Data to A Bus
L L	H H	X H or L	X X	L H	X X	Input	Output	Real Time $\overline{A}$ Data to B Bus Stored $\overline{A}$ Data to B Bus

H = High Level L = Low Level X = Irrelevant ↑ = Low-to-high level transition
*The data output functions may be enabled or disabled by various signals at the $\overline{G}$ and DIR inputs. Data input functions are always enabled, i.e., data at the bus pins will be stored on every low-to-high transition on the clock inputs.

FIGURE 2. Truth tables - Continued.

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Device types 01, 02 and 03

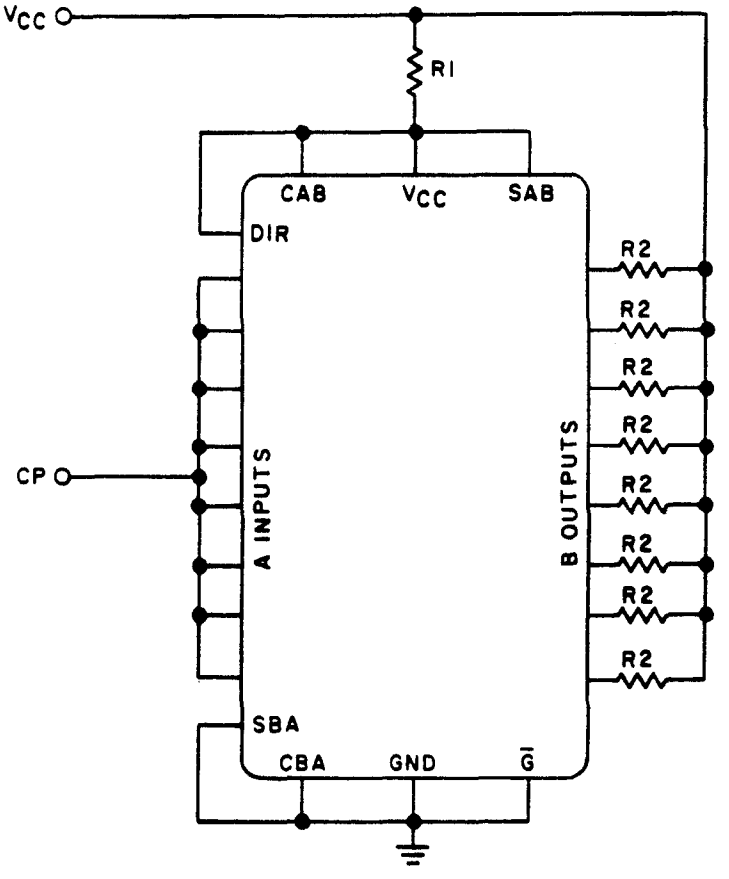


NOTES:
1. V_{CC} and R1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 3. Burn-in and life test circuit.

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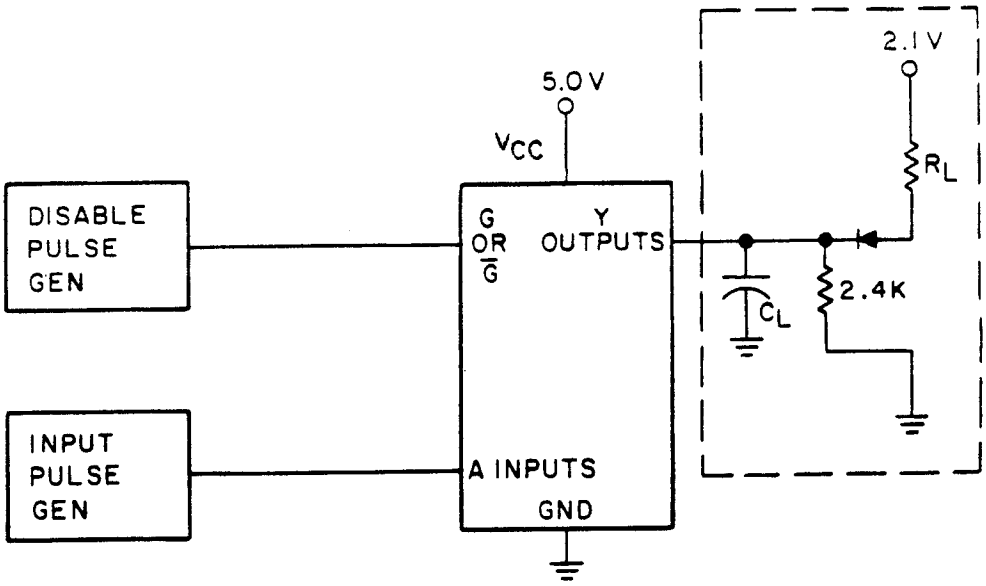
Device types 04 and 05



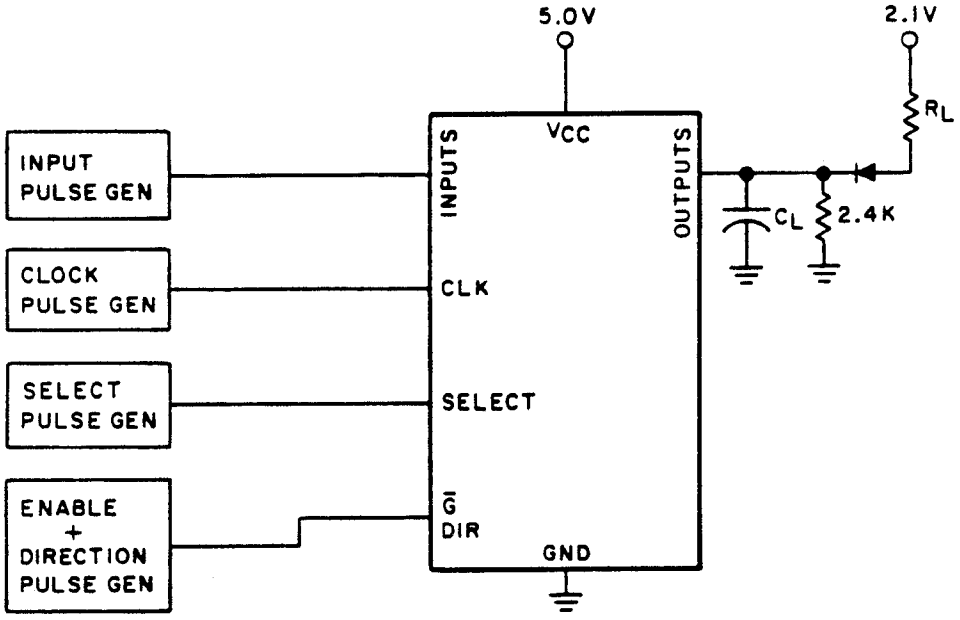
NOTES:
1. V_{CC} and R1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.5 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 3. Burn-in and life test circuit - Continued.

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TEST CIRCUIT TYPES 01, 02 and 03

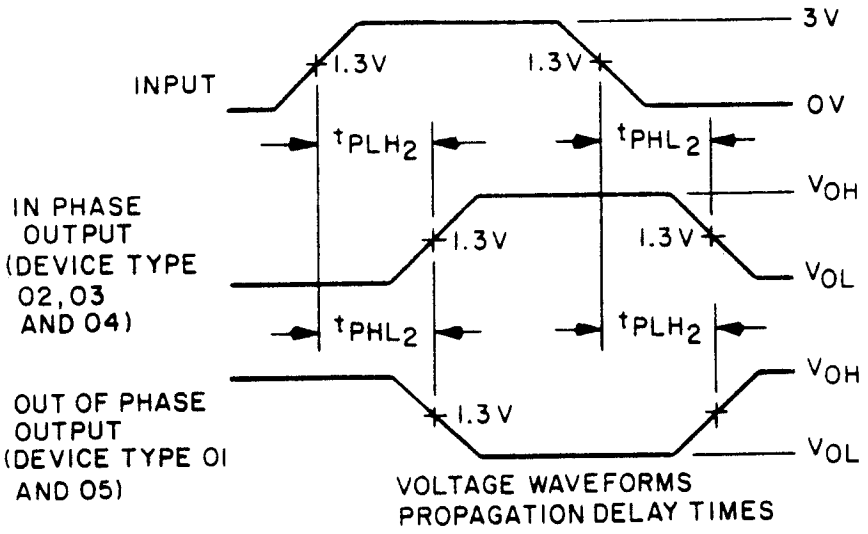


TEST CIRCUIT TYPES 04 and 05

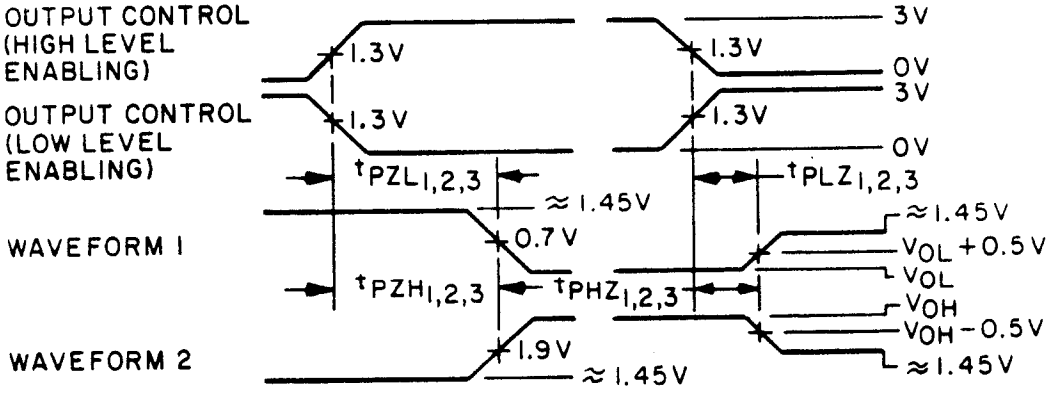
NOTES:
1. R_L = 110Ω ±5%.
2. All diodes are 1N3064 or equivalent.
3. C_L = 50 pF ±10% including probe and jig capacitance.
4. The pulse generators have the following characteristics: V_{gen} = 3 V, PRR ≤ 1 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, Z_{OUT} = 50Ω.
5. Clock pulse characteristics: t_p (CLK) = 20 ns, t_{setup} = 20 ns.

FIGURE 4. Switching time test circuit and waveforms.

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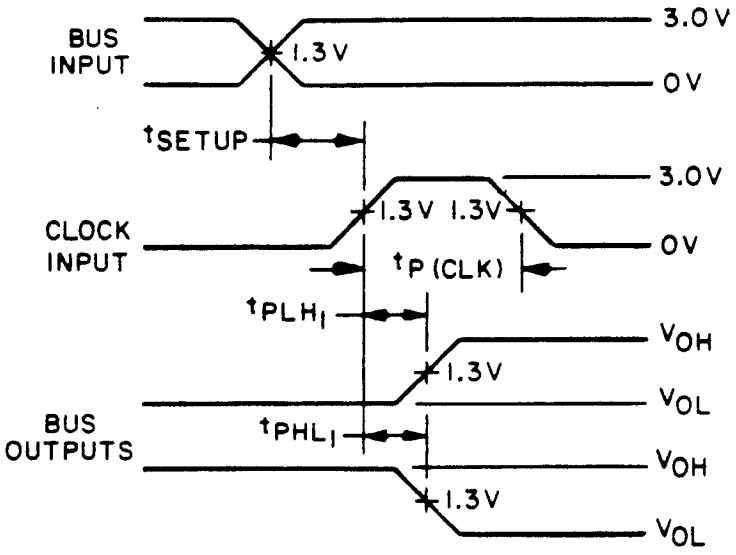
VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



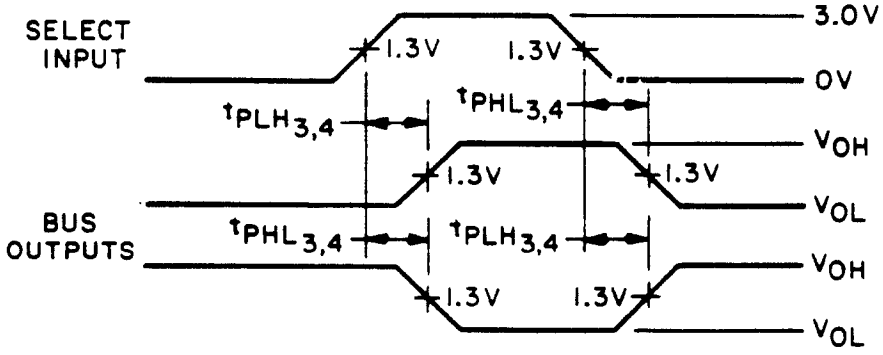
VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, THREE-STATE OUTPUTS, ALL DEVICES

FIGURE 4. Switching time test circuit and waveforms - Continued.

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CLOCK TO OUTPUT (TYPES 04 & 05)



SELECT TO OUTPUTS (TYPES 04 & 05)

FIGURE 4. Switching time test circuit and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/ 2 Test no.	1 2 CAB	2 3 NC	3 4 1A	4 6 2A	5 8 3A	6 9 4A	7 10 GND	8 12 4B	9 13 3B	10 14 2B	11 16 1B	12 18 NC	13 19 GBA	14 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	0.7 V		0.7 V				GND			-3 mA	-3 mA		0.7 V	4.5 V	1B	2.4		V
			2	"		"	0.7 V			"			"			"	"	2B	"		"
			3	"				0.7 V		"		-3 mA				"	"	3B	"		"
			4	"					0.7 V	"	-3 mA					"	"	4B	"		"
			5	2.0 V		-3 mA				"				0.7 V		2.0 V	"	1A	"		"
			6	"			-3 mA			"			0.7 V			"	"	2A	"		"
			7	"				-3 mA		"		0.7 V				"	"	3A	"		"
			8	"					-3 mA	"	0.7 V					"	"	4A	"		"
	V_{OH2}	3006	9	0.5 V		0.5 V				"			-12 mA	-12 mA		0.5 V	4.5 V	1B	2.0		V
			10	"		"	0.5 V			"			"			"	"	2B	"		"
			11	"				0.5 V		"		-12 mA				"	"	3B	"		"
			12	"					0.5 V	"	-12 mA					"	"	4B	"		"
			13	2.0 V		-12 mA				"				0.5 V		2.0 V	"	1A	"		"
			14	"			-12 mA			"			0.5 V			"	"	2A	"		"
			15	"				-12 mA		"		0.5 V				"	"	3A	"		"
			16	"					-12 mA	"	0.5 V					"	"	4A	"		"
	V_{OL}	3007	17	0.7 V		2.0 V				"			12 mA	12 mA		0.7 V	"	1B		0.4	"
			18	"		"	2.0 V			"			"			"	"	2B		"	"
			19	"				2.0 V		"		12 mA				"	"	3B		"	"
			20	"					2.0 V	"	12 mA					"	"	4B		"	"
			21	2.0 V		12 mA				"				2.0 V		2.0 V	"	1A		"	"
			22	"			12 mA			"			2.0 V			"	"	2A		"	"
			23	"				12 mA		"		2.0 V				"	"	3A		"	"
			24	"					12 mA	"	2.0 V					"	"	4A		"	"
	I_{OZH}		25			2.7 V				"						0.7 V	5.5 V	1A		2/	2/
			26				2.7 V			"						"	"	2A		"	"
			27					2.7 V		"						"	"	3A		"	"
			28						2.7 V	"						"	"	4A		"	"
			29	2.0 V						"	2.7 V					"	"	4B		"	"
			30	"						"		2.7 V				"	"	3B		"	"
			31	"						"			2.7 V			"	"	2B		"	"
			32	"						"				2.7 V		"	"	1B		"	"
	I_{OZL}		33			0.4 V				"						0.7 V	"	1A		-200	μA
			34				0.4 V			"						"	"	2A		"	"
			35					0.4 V		"						"	"	3A		"	"
			36						0.4 V	"						"	"	4A		"	"
			37	2.0 V						"	0.4 V					"	"	4B		"	"
			38	"						"		0.4 V				"	"	3B		"	"
			39	"						"			0.4 V			"	"	2B		"	"
			40	"						"				0.4 V		"	"	1B		"	"
	I_{IL}	3009	41	0.4 V		0.4 V				"							"	CAB	3/	3/	"
			42	"		"				"						GND	"	1A	"	"	"
			43	"			0.4 V			"						"	"	2A	"	"	"
			44	"				0.4 V		"						"	"	3A	"	"	"
			45	"					0.4 V	"						"	"	4A	"	"	"
			46	5.5 V						"	0.4 V					"	"	4B	"	"	"
			47	"						"		0.4 V				"	"	3B	"	"	"
			48	"						"			0.4 V			"	"	2B	"	"	"
			49	"						"				0.4 V		"	"	1B	"	"	"
			50	"						"						0.4 V	"	GBA	"	"	"
	I_{IH1}	3010	51	2.7 V		2.7 V				"							"	CAB		20	"
			52	"		"				"						GND	"	1A		"	"
			53	"			2.7 V			"						"	"	2A		"	"
			54	"				2.7 V		"						"	"	3A		"	"
			55	"					2.7 V	"						"	"	4A		"	"
			56	5.5 V						"	2.7 V					"	"	4B		"	"
			57	"						"		2.7 V				"	"	3B		"	"
			58	"						"			2.7 V			"	"	2B		"	"
			59	"						"				2.7 V		"	"	1B		"	"
			60	"						"						2.7 V	"	GBA		"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	GAB	NC	1A	2A	3A	4A	GND	4B	3B	2B	1B	NC	GBA	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	61	5.5 V						GND							5.5 V	$\overline{GAB}$		100	µA
		"	62			5.5 V				"						GND	"	1A		"	"
		"	63				5.5 V			"						"	"	2A		"	"
		"	64					5.5 V		"						"	"	3A		"	"
		"	65						5.5 V	"						"	"	4A		"	"
		"	66	5.5 V						"	5.5 V						"	4B		"	"
		"	67	"						"		5.5 V					"	3B		"	"
		"	68	"						"			5.5 V				"	2B		"	"
		"	69	"						"				5.5 V			"	1B		"	"
		"	70							"						5.5 V	"	GBA		"	"
	V_{IC}		71	-18 mA						"							4.5 V	$\overline{GAB}$		-1.5	V
			72			-18 mA				"						GND	"	1A		"	"
			73				-18 mA			"						"	"	2A		"	"
			74					-18 mA		"						"	"	3A		"	"
			75						-18 mA	"						"	"	4A		"	"
			76	5.5 V						"	-18 mA						"	4B		"	"
			77	"						"		-18 mA					"	3B		"	"
			78	"						"			-18 mA				"	2B		"	"
			79	"						"				-18 mA			"	1B		"	"
			80							"						-18 mA	"	GBA		"	"
	I_{CCH}	3005	81	GND		GND	GND	GND	GND	"						GND	5.5 V	V_{CC}		38	mA
	I_{CCL}	3005	82	GND		5.5 V	5.5 V	5.5 V	5.5 V	"						GND	"	V_{CC}		50	"
	I_{CCZ}	3005	83	5.5 V						"						GND	"	V_{CC}		50	"
	I_{OS}	3011	84	GND		GND				"				GND		GND	"	1B	-40	-225	"
		"	85	"			GND			"			GND			"	"	2B		"	"
		"	86	"				GND		"		GND				"	"	3B		"	"
		"	87	"					GND	"	GND					"	"	4B		"	"
		"	88	4.5 V		GND				"				GND		4.5 V	"	1A		"	"
		"	89	"			GND			"			GND			"	"	2A		"	"
		"	90	"				GND		"		GND				"	"	3A		"	"
		"	91	"					GND	"	GND					"	"	4A		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Truth table tests 4/		92	B		A	A	A	A	GND	L	L	L	L		B	5.0 V	5/			
			93	B		B	B	B	B	"	H	H	H	H		B	"				
			94	A		L	L	L	L	"	A	A	A	A		A	"				
			95	A		H	H	H	H	"	B	B	B	B		A	"				
		Test 96 optional	96	A		H	H	H	H	"	H	H	H	H		B	"	6/ (for test 96 only)			
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PLH2}	3003 See fig. 4	97	GND		IN				GND			OUT	OUT		GND	5.0 V	1A → 1B	2	19	ns
			98	"			IN			"			OUT			"	"	2A → 2B	"	"	"
			99	"				IN		"		OUT				"	"	3A → 3B	"	"	"
			100	"					IN	"	OUT					"	"	4A → 4B	"	"	"
			101	4.5 V		OUT				"				IN		4.5 V	"	1B → 1A	"	"	"
			102	"			OUT			"			IN			"	"	2B → 2A	"	"	"
			103	"				OUT		"		IN				"	"	3B → 3A	"	"	"
			104	"					OUT	"	IN					"	"	4B → 4A	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/ 2 Test no.	1 2 GAB	2 3 NC	3 4 1A	4 6 2A	5 8 3A	6 9 4A	7 10 GND	8 12 4B	9 13 3B	10 14 2B	11 16 1B	12 18 NC	13 19 GBA	14 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9	t_{PHL2}	3003	105	GND		IN				GND				OUT		GND	5.0 V	1A > 1B	2	23	ns
T_C = +25°C		See fig.	106	"			IN			"			OUT			"	"	2A > 2B	"	"	"
		4 *	107	"				IN		"		OUT				"	"	3A > 3B	"	"	"
		"	108	"					IN	"	OUT					"	"	4A > 4B	"	"	"
		"	109	4.5 V		OUT				"				IN		4.5 V	"	1B > 1A	"	"	"
		"	110	"			OUT			"			IN				"	2B > 2A	"	"	"
		"	111	"				OUT		"		IN					"	3B > 3A	"	"	"
		"	112	"					OUT	"	IN						"	4B > 4A	"	"	"
	t_{PZL1}	"	113	IN		4.5 V				"				OUT		GND	"	$\overline{GAB}$ > 1B	"	35	"
		"	114	"			4.5 V			"			OUT			"	"	$\overline{GAB}$ > 2B	"	"	"
		"	115	"				4.5 V		"		OUT				"	"	$\overline{GAB}$ > 3B	"	"	"
		"	116	"					4.5 V	"	OUT					"	"	$\overline{GAB}$ > 4B	"	"	"
		"	117	4.5 V		OUT				"				4.5 V		IN	"	GBA > 1A	"	"	"
		"	118	"			OUT			"			4.5 V			"	"	GBA > 2A	"	"	"
		"	119	"				OUT		"		4.5 V				"	"	GBA > 3A	"	"	"
		"	120	"					OUT	"	4.5 V					"	"	GBA > 4A	"	"	"
	t_{PZH1}	"	121	IN		GND				"				OUT		GND	"	$\overline{GAB}$ > 1B	"	28	"
		"	122	"			GND			"			OUT			"	"	$\overline{GAB}$ > 2B	"	"	"
		"	123	"				GND		"		OUT				"	"	$\overline{GAB}$ > 3B	"	"	"
		"	124	"					GND	"	OUT					"	"	$\overline{GAB}$ > 4B	"	"	"
		"	125	4.5 V		OUT				"				GND		IN	"	GBA > 1A	"	"	"
		"	126	"			OUT			"			GND			"	"	GBA > 2A	"	"	"
		"	127	"				OUT		"		GND				"	"	GBA > 3A	"	"	"
		"	128	"					OUT	"	GND					"	"	GBA > 4A	"	"	"
	t_{PLZ1}	"	129	IN		4.5 V				"				OUT		GND	"	$\overline{GAB}$ > 1B	"	30	"
		"	130	"			4.5 V			"			OUT			"	"	$\overline{GAB}$ > 2B	"	"	"
		"	131	"				4.5 V		"		OUT				"	"	$\overline{GAB}$ > 3B	"	"	"
		"	132	"					4.5 V	"	OUT					"	"	$\overline{GAB}$ > 4B	"	"	"
		"	133	4.5 V		OUT				"				4.5 V		IN	"	GBA > 1A	"	"	"
		"	134	"			OUT			"			4.5 V			"	"	GBA > 2A	"	"	"
		"	135	"				OUT		"		4.5 V				"	"	GBA > 3A	"	"	"
		"	136	"					OUT	"	4.5 V					"	"	GBA > 4A	"	"	"
	t_{PHZ1}	"	137	IN		GND				"				OUT		GND	"	$\overline{GAB}$ > 1B	"	35	"
		"	138	"			GND			"			OUT			"	"	$\overline{GAB}$ > 2B	"	"	"
		"	139	"				GND		"		OUT				"	"	$\overline{GAB}$ > 3B	"	"	"
		"	140	"					GND	"	OUT					"	"	$\overline{GAB}$ > 4B	"	"	"
		"	141	4.5 V		OUT				"				GND		IN	"	GBA > 1A	"	"	"
		"	142	"			OUT			"			GND			"	"	GBA > 2A	"	"	"
		"	143	"				OUT		"		GND				"	"	GBA > 3A	"	"	"
		"	144	"					OUT	"	GND					"	"	GBA > 4A	"	"	"
10	t_{PLH2} t_{PHL2} t_{PZL1} t_{PZH1} t_{PLZ1} t_{PHZ1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	" " " " " "	25 30 46 36 39 46	" " " " " "
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are N/C.
2/ The I_{OZH} limit for circuits D and E shall be 20 µA maximum; the limit for circuits A, B, and C shall be 40 µA maximum.
3/ The I_{IL} limits are as follows:

Test	Min/Max limits µA for circuit: A	B	C	D	E
I_{IL}	-57/-200	0/-100	0/-200	-10/-150	0/-150

4/ A = 3.0 V minimum; B = 0.0 V or GND.
5/ H > 1.5 V; L < 1.5 V.
6/ Add resistor of 0.5 kΩ to 5 kΩ between V_{CC} and each output.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/ 2 Test no.	1 2 GAB	2 3 NC	3 4 1A	4 6 2A	5 8 3A	6 9 4A	7 10 GND	8 12 4B	9 13 3B	10 14 2B	11 16 1B	12 18 NC	13 19 GBA	14 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	0.7 V		2.0 V				GND				-3 mA		0.7 V	4.5 V	1B	2.4		V
		"	2	"			2.0 V			"			-3 mA			"	"	2B	"		"
		"	3	"				2.0 V		"		-3 mA				"	"	3B	"		"
		"	4	"					2.0 V	"	-3 mA					"	"	4B	"		"
		"	5	2.0 V		-3 mA				"				2.0 V		2.0 V	"	1A	"		"
		"	6	"			-3 mA			"			2.0 V			"	"	2A	"		"
		"	7	"				-3 mA		"		2.0 V				"	"	3A	"		"
		"	8	"					-3 mA	"	2.0 V					"	"	4A	"		"
	V_{OH2}	3006	9	0.5 V		2.0 V				"				-12 mA		0.5 V	"	1B	2.0		"
		"	10	"			2.0 V			"			-12 mA			"	"	2B	"		"
		"	11	"				2.0 V		"		-12 mA				"	"	3B	"		"
		"	12	"					2.0 V	"	-12 mA					"	"	4B	"		"
		"	13	2.0 V		-12 mA				"				2.0 V		2.0 V	"	1A	"		"
		"	14	"			-12 mA			"			2.0 V			"	"	2A	"		"
		"	15	"				-12 mA		"		2.0 V				"	"	3A	"		"
		"	16	"					-12 mA	"	2.0 V					"	"	4A	"		"
	V_{OL}	3007	17	0.7 V		0.7 V				"				12 mA		0.7 V	"	1B		0.4	"
		"	18	"			0.7 V			"			12 mA			"	"	2B		"	"
		"	19	"				0.7 V		"		12 mA				"	"	3B		"	"
		"	20	"					0.7 V	"	12 mA					"	"	4B		"	"
		"	21	2.0 V		12 mA				"				0.7 V		2.0 V	"	1A		"	"
		"	22	"			12 mA			"			0.7 V			"	"	2A		"	"
		"	23	"				12 mA		"		0.7 V				"	"	3A		"	"
		"	24	"					12 mA	"	0.7 V					"	"	4A		"	"
	I_{OZH}		25			2.7 V				"						0.7 V	5.5 V	1A		2/	2/
			26				2.7 V			"						"	"	2A		"	"
			27					2.7 V		"						"	"	3A		"	"
			28						2.7 V	"						"	"	4A		"	"
			29	2.0 V						"	2.7 V					"	"	4B		"	"
			30	"						"		2.7 V				"	"	3B		"	"
			31	"						"			2.7 V			"	"	2B		"	"
			32	"						"				2.7 V		"	"	1B		"	"
	I_{OZL}		33			0.4 V				"						0.7 V	"	1A		-200	μA
			34				0.4 V			"						"	"	2A		"	"
			35					0.4 V		"						"	"	3A		"	"
			36						0.4 V	"						"	"	4A		"	"
			37	2.0 V						"	0.4 V					"	"	4B		"	"
			38	"						"		0.4 V				"	"	3B		"	"
			39	"						"			0 4 V			"	"	2B		"	"
			40	"						"				0.4 V		"	"	1B		"	"
	I_{IL}	3009	41	0.4 V						"							"	GAB	3/	3/	"
		"	42	"		0.4 V				"						GND	"	1A	"	"	"
		"	43	"			0.4 V			"						"	"	2A	"	"	"
		"	44	"				0.4 V		"						"	"	3A	"	"	"
		"	45	"					0.4 V	"						"	"	4A	"	"	"
		"	46	5.5 V						"	0.4 V						"	4B	"	"	"
		"	47	"						"		0.4 V					"	3B	"	"	"
		"	48	"						"			0.4 V				"	2B	"	"	"
		"	49	"						"				0.4 V			"	1B	"	"	"
		"	50	"						"						0.4 V	"	GBA	"	"	"
	I_{IH1}	3010	51	2.7 V						"							"	GAB		20	"
		"	52	"		2.7 V				"						GND	"	1A		"	"
		"	53	"			2.7 V			"						"	"	2A		"	"
		"	54	"				2.7 V		"						"	"	3A		"	"
		"	55	"					2.7 V	"						"	"	4A		"	"
		"	56	5.5 V						"	2.7 V						"	4B		"	"
		"	57	"						"		2.7 V					"	3B		"	"
		"	58	"						"			2.7 V				"	2B		"	"
		"	59	"						"				2.7 V			"	1B		"	"
		"	60	"						"						2.7 V	"	GBA		"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	$\overline{GAB}$	NC	1A	2A	3A	4A	GND	4B	3B	2B	1B	NC	GBA	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	61	5.5 V						GND							5.5 V	$\overline{GAB}$		100	µA
		"	62			5.5 V				"						GND	"	1A		"	"
		"	63				5.5 V			"						"	"	2A		"	"
		"	64					5.5 V		"						"	"	3A		"	"
		"	65						5.5 V	"						"	"	4A		"	"
		"	66	5.5 V						"	5.5 V						"	4B		"	"
		"	67	"						"		5.5 V					"	3B		"	"
		"	68	"						"			5.5 V				"	2B		"	"
		"	69	"						"				5.5 V			"	1B		"	"
		"	70							"						5.5 V	"	GBA		"	"
	V_{IC}		71	-18 mA						"							4.5 V	$\overline{GAB}$		-1.5	V
			72			-18 mA				"						GND	"	1A		"	"
			73				-18 mA			"						"	"	2A		"	"
			74					-18 mA		"						"	"	3A		"	"
			75						-18 mA	"						"	"	4A		"	"
			76	5.5 V						"	-18 mA					"	"	4B		"	"
			77	"						"		-18 mA					"	3B		"	"
			78	"						"			-18 mA				"	2B		"	"
			79	"						"				-18 mA			"	1B		"	"
			80	"						"						-18 mA	"	GBA		"	"
	I_{CCH}	3005	81	GND		5.5 V	5.5 V	5.5 V	5.5 V	"						GND	5.5 V	V_{CC}		38	mA
	I_{CCL}	3005	82	GND		GND	GND	GND	GND	"						GND	"	V_{CC}		50	"
	I_{CCZ}	3005	83	5.5 V						"						GND	"	V_{CC}		50	"
	I_{OS}	3011	84	GND		5.5 V				"				GND		GND	"	1B	-40	-225	"
		"	85	"			5.5 V			"			GND			"	"	2B	"	"	"
		"	86	"				5.5 V		"		GND				"	"	3B	"	"	"
		"	87	"					5.5 V	"	GND					"	"	4B	"	"	"
		"	88	5.5 V		GND				"				5.5 V		5.5 V	"	1A	"	"	"
		"	89	"			GND			"			5.5 V			"	"	2A	"	"	"
		"	90	"				GND		"		5.5 V				"	"	3A	"	"	"
		"	91	"					GND	"	5.5 V					"	"	4A	"	"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Truth table tests 4/		92	B		A	A	A	A	GND	H	H	H	H		B	5.0 V				
			93	B		B	B	B	B	"	L	L	L	L		B	"				
			94	A		H	H	H	H	"	A	A	A	A		A	"	5/			
			95	A		L	L	L	L	"	B	B	B	B		A	"				
			96	A		H	H	H	H	"	H	H	H	H		B	"	6/ (test 96 only)			
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PLH2}	3003 See fig. 4	97	GND		IN				GND				OUT		GND	5.0 V	1A > 1B	2	19	ns
		"	98	"			IN			"			OUT			"	"	2A > 2B	"	"	"
		"	99	"				IN		"		OUT				"	"	3A > 3B	"	"	"
		"	100	"					IN	"	OUT					"	"	4A > 4B	"	"	"
		"	101	4.5 V		OUT				"				IN		4.5 V	"	1B > 1A	"	"	"
		"	102	"			OUT			"			IN			"	"	2B > 2A	"	"	"
		"	103	"				OUT		"		IN				"	"	3B > 3A	"	"	"
		"	104	"					OUT	"	IN					"	"	4B > 4A	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, C, D Case 1/2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	GAB	NC	IA	2A	3A	4A	GND	4B	3B	2B	1B	NC	GBA	V_{CC}				
9	t_{PHL2}	3003	105	GND		IN				GND				OUT		GND	5.0 V	1A › 1B	2	23	ns
T_C = +25°C		See fig.	106	"			IN			"			OUT			"	"	2A › 2B	"	"	"
		4	107	"				IN		"		OUT				"	"	3A › 3B	"	"	"
		"	108	"					IN	"	OUT					"	"	4A › 4B	"	"	"
		"	109	4.5 V		OUT				"				IN		4.5 V	"	1B › 1A	"	"	"
		"	110	"			OUT			"			IN			"	"	2B › 2A	"	"	"
		"	111	"				OUT		"		IN				"	"	3B › 3A	"	"	"
		"	112	"					OUT	"	IN					"	"	4B › 4A	"	"	"
	t_{PZL1}	"	113	IN		GND				"				OUT		GND	"	GAB › 1B	"	35	"
		"	114	"			GND			"			OUT			"	"	GAB › 2B	"	"	"
		"	115	"				GND		"		OUT				"	"	GAB › 3B	"	"	"
		"	116	"					GND	"	OUT					"	"	GAB › 4B	"	"	"
		"	117	4.5 V		OUT				"				GND		IN	"	GBA › 1A	"	"	"
		"	118	"			OUT			"			GND			"	"	GBA › 2A	"	"	"
		"	119	"				OUT		"		GND				"	"	GBA › 3A	"	"	"
		"	120	"					OUT	"	GND					"	"	GBA › 4A	"	"	"
	t_{PZH1}	"	121	IN		4.5 V				"				OUT		GND	"	GAB › 1B	"	28	"
		"	122	"			4.5 V			"			OUT			"	"	GAB › 2B	"	"	"
		"	123	"				4.5 V		"		OUT				"	"	GAB › 3B	"	"	"
		"	124	"					4.5 V	"	OUT					"	"	GAB › 4B	"	"	"
		"	125	4.5 V		OUT				"				4.5 V		IN	"	GBA › 1A	"	"	"
		"	126	"			OUT			"			4.5 V			"	"	GBA › 2A	"	"	"
		"	127	"				OUT		"		4.5 V				"	"	GBA › 3A	"	"	"
		"	128	"					OUT	"	4.5 V					"	"	GBA › 4A	"	"	"
	t_{PLZ1}	"	129	IN		GND				"				OUT		GND	"	GAB › 1B	"	30	"
		"	130	"			GND			"			OUT			"	"	GAB › 2B	"	"	"
		"	131	"				GND		"		OUT				"	"	GAB › 3B	"	"	"
		"	132	"					GND	"	OUT					"	"	GAB › 4B	"	"	"
		"	133	4.5 V		OUT				"				GND		IN	"	GBA › 1A	"	"	"
		"	134	"			OUT			"			GND			"	"	GBA › 2A	"	"	"
		"	135	"				OUT		"		GND				"	"	GBA › 3A	"	"	"
		"	136	"					OUT	"	GND					"	"	GBA › 4A	"	"	"
	t_{PHZ1}	"	137	IN		4.5 V				"				OUT		GND	"	GAB › 1B	"	35	"
		"	138	"			4.5 V			"			OUT			"	"	GAB › 2B	"	"	"
		"	139	"				4.5 V		"		OUT				"	"	GAB › 3B	"	"	"
		"	140	"					4.5 V	"	OUT					"	"	GAB › 4B	"	"	"
		"	141	4.5 V		OUT				"				4.5 V		IN	"	GBA › 1A	"	"	"
		"	142	"			OUT			"			4.5 V			"	"	GBA › 2A	"	"	"
		"	143	"				OUT		"		4.5 V				"	"	GBA › 3A	"	"	"
		"	144	"					OUT	"	4.5 V					"	"	GBA › 4A	"	"	"
10	t_{PLH2} t_{PHL2} t_{PZL1} t_{PZH1} t_{PLZ1} t_{PHZ1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	"	25	"
																			"	30	"
																			"	46	"
																			"	36	"
																			"	39	"
																			"	46	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are N/C.
2/ The I_{OZH} limit for circuit D and E shall be 20 µA maximum; the limit for circuits A, B, and C shall be 40 µA maximum.
3/ I_{IL} limits are as follows:

Test	Min/Max limits µA for circuit: A	B	C	D	E
I_{IL}	-5/-200	0/-100	0/-200	-10/-150	0/-150

4/ A = 3.0 V minimum; B = 0.0 V or GND.
5/ H ≥ 1.5 V, L ≤ 1.5 V.
6/ Add resistor of 0.5 kΩ to 5 kΩ between V_{CC} and each output.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R and S Test no.	1 DIR	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 A_6	8 A_7	9 A_8	10 GND	11 B_8	12 B_7	13 B_6	14 B_5	15 B_4	16 B_3	17 B_2	18 B_1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	0.7 V	4.5 V	B_1	2.4		V
			2	"	"								"									"	"	B_2	"		"
			3	"	"								"									"	"	B_3	"		"
			4	"	"								"									"	"	B_4	"		"
			5	"	"								"									"	"	B_5	"		"
			6	"	"								"									"	"	B_6	"		"
			7	"	"								"									"	"	B_7	"		"
			8	"	"								"									"	"	B_8	"		"
			9	0.7 V	-3 mA	-3 mA							"								2.0 V	"	"	A_1	"		"
			10	"	"								"						2.0 V	2.0 V		"	"	A_2	"		"
			11	"	"		-3 mA						"					2.0 V				"	"	A_3	"		"
			12	"	"			-3 mA					"									"	"	A_4	"		"
			13	"	"				-3 mA				"				2.0 V					"	"	A_5	"		"
			14	"	"					-3 mA			"			2.0 V						"	"	A_6	"		"
			15	"	"						-3 mA		"		2.0 V							"	"	A_7	"		"
			16	"	"							-3 mA	"	2.0 V								"	"	A_8	"		"
	V_{OH2}	3006	17	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	0.5 V	"	B_1	2.0		"
			18	"	"								"									"	"	B_2	"		"
			19	"	"								"									"	"	B_3	"		"
			20	"	"								"									"	"	B_4	"		"
			21	"	"								"									"	"	B_5	"		"
			22	"	"								"									"	"	B_6	"		"
			23	"	"								"		-12 mA							"	"	B_7	"		"
			24	"	"								"									"	"	B_8	"		"
			25	0.5 V	-12 mA	-12 mA							"	-12 mA							2.0 V	"	"	A_1	"		"
			26	"	"								"						2.0 V	2.0 V		"	"	A_2	"		"
			27	"	"		-12 mA						"					2.0 V				"	"	A_3	"		"
			28	"	"			-12 mA					"									"	"	A_4	"		"
			29	"	"				-12 mA				"				2.0 V					"	"	A_5	"		"
			30	"	"					-12 mA			"			2.0 V						"	"	A_6	"		"
			31	"	"						-12 mA		"		2.0 V							"	"	A_7	"		"
			32	"	"							-12 mA	"	2.0 V								"	"	A_8	"		"
	V_{OL}	3007	33	2.0 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	"	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	0.7 V	"	B_1		0.4	"
			34	"	"								"									"	"	B_2	"		"
			35	"	"								"									"	"	B_3	"		"
			36	"	"								"									"	"	B_4	"		"
			37	"	"								"									"	"	B_5	"		"
			38	"	"								"									"	"	B_6	"		"
			39	"	"								"									"	"	B_7	"		"
			40	"	"								"									"	"	B_8	"		"
			41	0.7 V	12 mA	12 mA							"								0.7 V	"	"	A_1	"		"
			42	"	"		12 mA						"						0.7 V	0.7 V		"	"	A_2	"		"
			43	"	"			12 mA					"					0.7 V				"	"	A_3	"		"
			44	"	"				12 mA				"				0.7 V					"	"	A_4	"		"
			45	"	"					12 mA			"			0.7 V						"	"	A_5	"		"
			46	"	"						12 mA		"		0.7 V							"	"	A_6	"		"
			47	"	"							12 mA	"	0.7 V								"	"	A_7	"		"
			48	"	"								"									"	"	A_8	"		"
	V_{IC}		49	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	5.5 V	"	DIR		-1.5	"
			50	"	"								"									"	"	A_1	"		"
			51	"	"								"									"	"	A_2	"		"
			52	"	"								"									"	"	A_3	"		"
			53	"	"								"									"	"	A_4	"		"
			54	"	"								"									"	"	A_5	"		"
			55	"	"								"									"	"	A_6	"		"
			56	"	"								"									"	"	A_7	"		"
			57	"	"								"									"	"	A_8	"		"
			58	"	"								"									"	"	B_8	"		"
			59	"	"								"									"	"	B_7	"		"
			60	"	"								"									"	"	B_6	"		"
			61	"	"								"									"	"	B_5	"		"
			62	"	"								"									"	"	B_4	"		"
			63	"	"								"									"	"	B_3	"		"
			64	"	"								"									"	"	B_2	"		"
			65	"	"								"									"	"	B_1	"		"
			66	"	"								"									"	"	G	"		"

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TABLE III. Group A inspection for device type 03 -Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases Z, R and S Test no.	1 DIR	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 A_6	8 A_7	9 A_8	10 GND	11 B_8	12 B_7	13 B_6	14 B_5	15 B_4	16 B_3	17 B_2	18 B_1	19 G	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{OZH}		67		2.7 V	2.7 V							GND									2.0 V	5.5 V	A_1		10 1/	µA
			68				2.7 V																	A_2			
			69					2.7 V																A_3			
			70						2.7 V															A_4			
			71							2.7 V														A_5			
			72								2.7 V													A_6			
			73									2.7 V												A_7			
			74											2.7 V										A_8			
			75												2.7 V									B_7			
			76													2.7 V								B_6			
			77														2.7 V							B_5			
			78															2.7 V						B_4			
			79																2.7 V					B_3			
			80																	2.7 V				B_2			
			81																		2.7 V			B_1			
			82																					G			
	I_{OZL}		83		0.4 V	0.4 V																		A_1		-200	
			84				0.4 V																	A_2			
			85					0.4 V																A_3			
			86						0.4 V															A_4			
			87							0.4 V														A_5			
			88								0.4 V													A_6			
			89									0.4 V												A_7			
			90											0.4 V										A_8			
			91												0.4 V									B_7			
			92													0.4 V								B_6			
			93														0.4 V							B_5			
			94															0.4 V						B_4			
			95																0.4 V					B_3			
			96																	0.4 V				B_2			
			97																		0.4 V			B_1			
			98																					G			
	I_{IL}	3009	99	0.4 V	0.4 V	0.4 V																5.5 V		DIR	-5	-200 2/	
			100				0.4 V																	A_1			
			101					0.4 V																A_2			
			102						0.4 V															A_3			
			103							0.4 V														A_4			
			104								0.4 V													A_5			
			105									0.4 V												A_6			
			106											0.4 V										A_7			
			107												0.4 V									A_8			
			108													0.4 V								B_7			
			109														0.4 V							B_6			
			110															0.4 V						B_5			
			111																0.4 V					B_4			
			112																	0.4 V				B_3			
			113																		0.4 V			B_2			
			114																			0.4 V		B_1			
			115																					G			
			116																								
	I_{IH1}	3010	117	2.7 V	2.7 V	2.7 V																5.5 V		DIR		20	
			118				2.7 V																	A_1			
			119					2.7 V																A_2			
			120						2.7 V															A_3			
			121							2.7 V														A_4			
			122								2.7 V													A_5			
			123									2.7 V												A_6			
			124											2.7 V										A_7			
			125												2.7 V									A_8			
			126													2.7 V								B_7			
			127														2.7 V							B_6			
			128															2.7 V						B_5			
			129																2.7 V					B_4			
			130																	2.7 V				B_3			
			131																		2.7 V			B_2			
			132																			2.7 V		B_1			
			133																					G			
			134																								

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TABLE III. Group A inspection for device type 03 -Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, 3, and 5 Test no.	1 DIR	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 A_6	8 A_7	9 A_8	10 GND	11 B_8	12 B_7	13 B_6	14 B_5	15 B_4	16 B_3	17 B_2	18 B_1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IN2}	3010	135	5.5 V									GND									5.5 V	5.5 V	DIR		100	µA
		"	136		5.5 V								"									"	"	A_1		"	"
		"	137			5.5 V							"									"	"	A_2		"	"
		"	138				5.5 V						"									"	"	A_3		"	"
		"	139					5.5 V					"									"	"	A_4		"	"
		"	140						5.5 V				"									"	"	A_5		"	"
		"	141							5.5 V			"									"	"	A_6		"	"
		"	142								5.5 V		"									"	"	A_7		"	"
		"	143									5.5 V	"									"	"	A_8		"	"
		"	144										"	5.5 V								"	"	B_8		"	"
		"	145										"		5.5 V							"	"	B_7		"	"
		"	146										"			5.5 V						"	"	B_6		"	"
		"	147										"				5.5 V					"	"	B_5		"	"
		"	148										"					5.5 V				"	"	B_4		"	"
		"	149										"						5.5 V			"	"	B_3		"	"
		"	150										"							5.5 V		"	"	B_2		"	"
		"	151										"									"	"	B_1		"	"
		"	152										"								5.5 V	"	"	$\overline{G}$		"	"
	I_{OS}	3011	153	5.5 V	5.5 V	5.5 V							"								GND	GND	"	B_1	-40	-225	mA
		"	154	"			5.5 V						"							GND	"	"	"	B_2	"	"	"
		"	155	"				5.5 V					"						GND		"	"	"	B_3	"	"	"
		"	156	"					5.5 V				"					GND			"	"	"	B_4	"	"	"
		"	157	"						5.5 V			"				GND				"	"	"	B_5	"	"	"
		"	158	"							5.5 V		"				GND					"	"	"	B_6	"	"	"
		"	159	"								5.5 V	"			GND						"	"	"	B_7	"	"	"
		"	160	"									"	GND							5.5 V	"	"	"	B_8	"	"	"
		"	161	GND	GND								"		GND							"	"	"	A_1	"	"	"
		"	162	"		GND							"			GND						"	"	"	A_2	"	"	"
		"	163	"			GND						"				GND					"	"	"	A_3	"	"	"
		"	164	"				GND					"					5.5 V				"	"	"	A_4	"	"	"
		"	165	"					GND				"						5.5 V			"	"	"	A_5	"	"	"
		"	166	"						GND			"			5.5 V						"	"	"	A_6	"	"	"
		"	167	"							GND		"							5.5 V		"	"	"	A_7	"	"	"
		"	168	"								GND	"	5.5 V								"	"	"	A_8	"	"	"
	I_{CCH}	3005	169	"									"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	V_{CC}		70	"
	I_{CCL}	3005	170	"									"	GND	GND	GND	GND	GND	GND	GND	GND	5.5 V	"	V_{CC}		90	"
	I_{CCZ}	3005	171										"										"	V_{CC}		95	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = +125°C.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except V_{IC} tests are omitted and T_C = -55°C.																										
7 3/ T_C = +25°C	Truth table tests		172	A	A	A	A	A	A	A	A	A	GND	H	H	H	H	H	H	H	H	B	5.0 V			4/	
			173	A	B	B	B	B	B	B	B	B	"	L	L	L	L	L	L	L	L	B	"				
			174	B	H	H	H	H	H	H	H	H	"	A	A	A	A	A	A	A	A	B	"				
			175	B	L	L	L	L	L	L	H	L	"	B	B	B	B	B	B	B	B	B	"				
			176	B	H	H	H	H	H	H	L	H	"	H	H	H	H	H	H	H	H	A	"			5/ (test 176 only)	
8	Same tests and terminal conditions as for subgroup 7, except T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PHL2}	3003	177	GND	OUT								GND									GND	5.0 V	B_1 > A_1	2	17	ns
		See fig. 4	178	"		OUT							"							IN	IN	"	"	B_2 > A_2	"	"	"
		"	179	"			OUT						"						IN			"	"	B_3 > A_3	"	"	"
		"	180	"				OUT					"					IN				"	"	B_4 > A_4	"	"	"
		"	181	"					OUT				"				IN					"	"	B_5 > A_5	"	"	"
		"	182	"						OUT			"			IN						"	"	B_6 > A_6	"	"	"
		"	183	"							OUT		"		IN							"	"	B_7 > A_7	"	"	"
		"	184	"								OUT	"	IN								"	"	B_8 > A_8	"	"	"
		"	185	4.5 V	IN								"							OUT		"	"	A_1 > B_1	"	"	"
		"	186	"		IN							"		IN				OUT			"	"	A_2 > B_2	"	"	"
		"	187	"			IN						"			IN						"	"	A_3 > B_3	"	"	"
		"	188	"				IN					"				IN					"	"	A_4 > B_4	"	"	"
		"	189	"					IN				"					IN				"	"	A_5 > B_5	"	"	"
		"	190	"						IN			"			OUT						"	"	A_6 > B_6	"	"	"
		"	191	"							IN		"		OUT							"	"	A_7 > B_7	"	"	"
		"	192	"								IN	"	OUT								"	"	A_8 > B_8	"	"	"

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, 3 IR and S Test no.	1 DIR	2 A_1	3 A_2	4 A_3	5 A_4	6 A_5	7 A_6	8 A_7	9 A_8	10 GND	11 B_8	12 B_7	13 B_6	14 B_5	15 B_4	16 B_3	17 B_2	18 B_1	19 $\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PLH2}	3003 (See fig. 4)	193	4.5 V	IIN	IIN	IIN	IIN	IIN	IIN	IIN	IIN	GND	IIN	IIN	IIN	IIN	IIN	IIN	IIN	IIN	GND	5.0 V	$A_1 \rightarrow B_1$ $A_2 \rightarrow B_2$ $A_3 \rightarrow B_3$ $A_4 \rightarrow B_4$ $A_5 \rightarrow B_5$ $A_6 \rightarrow B_6$ $A_7 \rightarrow B_7$ $A_8 \rightarrow B_8$ $B_1 \rightarrow A_1$ $B_2 \rightarrow A_2$ $B_3 \rightarrow A_3$ $B_4 \rightarrow A_4$ $B_5 \rightarrow A_5$ $B_6 \rightarrow A_6$ $B_7 \rightarrow A_7$ $B_8 \rightarrow A_8$	2	17	ns
	t_{PZL1}		209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 B_1 B_2 B_3 B_4 B_5 B_6 B_7 B_8	45		
	t_{PZH1}		225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	GND	4.5 V	A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 B_1 B_2 B_3 B_4 B_5 B_6 B_7 B_8	45		
	t_{PLZ1}		241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	GND	5.0 V	A_1 A_2 A_3 A_4 A_5 A_6 A_7 A_8 B_1 B_2 B_3 B_4 B_5 B_6 B_7 B_8	2	30	

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TABLE III. Group A inspection for device type 03 -Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases Z, R and S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	E	V_{CC}				
9 T_C = +25°C	t_{PHZ1}	3003 See fig. 4	257	4.5 V	4.5 V								GND										5.0 V	> B_1	2	30	ns
			258	"		4.5 V							"						OUT		OUT		"	> B_2	"	"	"
			259	"			4.5 V						"					OUT		OUT			"	> B_3	"	"	"
			260	"				4.5 V					"				OUT						"	> B_4	"	"	"
			261	"					4.5 V				"		OUT								"	> B_5	"	"	"
			262	"						4.5 V			"			OUT							"	> B_6	"	"	"
			263	"							4.5 V		"	OUT									"	> B_7	"	"	"
			264	"								4.5 V	"										"	> B_8	"	"	"
			265	GND	OUT								"										"	> A_1	"	"	"
			266	"		OUT							"						4.5 V		4.5 V		"	> A_2	"	"	"
			267	"			OUT						"							4.5 V			"	> A_3	"	"	"
			268	"				OUT					"					4.5 V					"	> A_4	"	"	"
			269	"					OUT				"				4.5 V						"	> A_5	"	"	"
			270	"						OUT			"			4.5 V							"	> A_6	"	"	"
			271	"							OUT		"		4.5 V								"	> A_7	"	"	"
			272	"								OUT	"										"	> A_8	"	"	"
10 T_C = +125°C	t_{PHL2}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																							"	22	"
	t_{PLH2}																								"	22	"
	t_{PZL1}																								"	58	"
	t_{PZH1}																								"	58	"
	t_{PLZ1}																								"	39	"
	t_{PHZ1}																								"	39	"
11 T_C=-55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																										

1/ The I_{OZH} limits for circuit E is +20 µA maximum.
2/ The I_{IL} limits for circuit E is 0 µA minimum, -150 µA maximum.
3/ A = 3.0 V minimum; B = 0.0 V or GND.
4/ H > 1.5 V; L < 1.5 V.
5/ Add resistor of 0.5 kΩ to 5 kΩ between V_{CC} and each output.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH1}	3006	1	0.5 V	0.5 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	0.5 V	0.5 V	0.5 V	4.5 V	B_1	2.4		V
			2																									B_2			
			3																									B_3			
			4																									B_4			
			5																									B_5			
			6																									B_6			
			7																									B_7			
			8																									B_8			
			9			0.5 V	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA		-3 mA	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V					A_1			
			10																									A_2			
			11																									A_3			
			12																									A_4			
			13																									A_5			
			14																									A_6			
			15																									A_7			
			16																									A_8			
	V_{OH2}	3006	17	"	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	"	"	"	"	B_1	2.0	-	"
			18																									B_2			
			19																									B_3			
			20																									B_4			
			21																									B_5			
			22																									B_6			
			23																									B_7			
			24																									B_8			
			25			0.5 V	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA		-12 mA	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V					A_1			
			26																									A_2			
			27																									A_3			
			28																									A_4			
			29																									A_5			
			30																									A_6			
			31																									A_7			
			32																									A_8			
	V_{OL}	3007	33	"	"	2.0 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	"	"	"	"	B_1		0.4	"
			34																									B_2			
			35																									B_3			
			36																									B_4			
			37																									B_5			
			38																									B_6			
			39																									B_7			
			40																									B_8			
			41			0.5 V	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA		12 mA	12 mA	12 mA	12 mA	0.5 V	0.5 V	0.5 V	0.5 V					A_1			
			42																									A_2			
			43																									A_3			
			44																									A_4			
			45																									A_5			
			46																									A_6			
			47																									A_7			
			48																									A_8			
	V_{IC}		49	-18 mA																								CAB		-1.5	"
			50		-18 mA																							SAB			
			51			-18 mA																						DIR			
			52				-18 mA																					A_1			
			53					-18 mA																				A_2			
			54						-18 mA																			A_3			
			55							-18 mA																		A_4			
			56								-18 mA																	A_5			
			57									-18 mA																A_6			
			58										-18 mA															A_7			
			59											-18 mA														A_8			
			60													-18 mA	-16 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA					B_8			
			61																									B_7			
			62																									B_6			
			63																									B_5			
			64																									B_4			
			65																									B_3			
			66																									B_2			
			67																									B_1			
			68																									$\overline{G}$			
			69																									SBA			
			70																									CBA			

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case L	Test no.	1 CLK AB	2 SEL AB	3 DIR	4 A_1	5 A_2	6 A_3	7 A_4	8 A_5	9 A_6	10 A_7	11 A_8	12 GND	13 B_8	14 B_7	15 B_6	16 B_5	17 B_4	18 B_3	19 B_2	20 B_1	21 $\overline{G}$	22 SEL BA	23 CLK BA	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
T_C = +25°C	I_{IL}	3009		71	0.4 V	0.4 V	0.4 V	0.4 V																			5.5 V	CAB	0	-200	µA
				72																								SAB			
				73					0.4 V																			DIR			
				74						0.4 V																		A_1			
				75							0.4 V																	A_2			
				76								0.4 V																A_3			
				77									0.4 V															A_4			
				78										0.4 V														A_5			
				79											0.4 V													A_6			
				80													0.4 V											A_7			
				81														0.4 V										A_8			
				82															0.4 V									B_8			
				83																0.4 V								B_7			
				84																	0.4 V							B_6			
				85																		0.4 V						B_5			
				86																			0.4 V					B_4			
				87																				0.4 V				B_3			
				88																					0.4 V			B_2			
				89																						0.4 V		B_1			
				90																							0.4 V	SBA			
				91																								CBA			
				92																											
	I_{IH1}	3010		93	2.7 V	2.7 V	2.7 V	2.7 V																			5.5 V	CAB	0	20	µA
				94																								SAB			
				95					2.7 V																			DIR			
				96						2.7 V																		A_1			
				97							2.7 V																	A_2			
				98								2.7 V																A_3			
				99									2.7 V															A_4			
				100										2.7 V														A_5			
				101											2.7 V													A_6			
				102													2.7 V											A_7			
				103														2.7 V										A_8			
				104															2.7 V									B_8			
				105																2.7 V								B_7			
				106																	2.7 V							B_6			
				107																		2.7 V						B_5			
				108																			2.7 V					B_4			
				109																				2.7 V				B_3			
				110																					2.7 V			B_2			
				111																						2.7 V		B_1			
				112																							2.7 V	SBA			
				113																								CBA			
				114																											
	I_{IH2}			115	5.5 V	5.5 V	5.5 V	5.5 V																			5.5 V	CAB	0	100	µA
				116																								SAB			
				117					5.5 V																			DIR			
				118						5.5 V																		A_1			
				119							5.5 V																	A_2			
				120								5.5 V																A_3			
				121									5.5 V															A_4			
				122										5.5 V														A_5			
				123											5.5 V													A_6			
				124													5.5 V											A_7			
				125														5.5 V										A_8			
				126															5.5 V									B_8			
				127																5.5 V								B_7			
				128																	5.5 V							B_6			
				129																		5.5 V						B_5			
				130																			5.5 V					B_4			
				131																				5.5 V				B_3			
				132																					5.5 V			B_2			
				133																						5.5 V		B_1			
				134																							5.5 V	$\overline{G}$			
				135																								SBA			
				136																								CBA			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}				
1 T_C = +25°C	I_{OZL}		137–152				0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	GND	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	2.0 V			5.5 V	A_1–A_8, B_8–B_1		20	µA
	I_{OZH}		153–168				2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V					A_1–A_8, B_8–B_1		400	
	I_{OS}	3011	169–184	GND	GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND	4.5 V, GND		GND, 0.5 V	GND, 0.5 V	GND, 0.5 V	GND, 0.5 V	GND, 0.5 V	GND, 0.5 V	GND, 0.5 V	5.5 V	GND	GND	GND		B_1–B_8, A_1–A_8	-40	-225	mA
	I_{CCH}	3005	185			4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V														V_{CC}		145	
	I_{CCL}		186			4.5 V	GND	GND	GND	GND	GND	GND	GND	GND														V_{CC}		185	
	I_{CCZ}		187			4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V										4.5 V				V_{CC}		185	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																														
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																														
7 T_C = +25°C	Truth table tests 1/		188–200	B, A, B	B, A, B	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	A, B, A	GND	H, L, L	H, L, L	H, L, L	H, L, L	H, L, L	H, L, L	H, L, L	H, L, L	B	B	B	4.5 V		2/ 3/		

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}				
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and T_C = -55°C																														
9 T_C = +25°C	t_{PLH1}	3003 (fig. 4)	201	IN	4.5 V	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	5.0 V	CAB > B_1	2	30	ns
			202																									CAB > B_2			
			203																									CAB > B_3			
			204																									CAB > B_4			
			205																									CAB > B_5			
			206																									CAB > B_6			
			207																									CAB > B_7			
			208																									CAB > B_8			
			209	GND	GND	GND	OUT	OUT																	4.5 V	IN		CBA > A_1			
			210																									CBA > A_2			
			211						OUT											IN								CBA > A_3			
			212							OUT											IN							CBA > A_4			
			213								OUT								IN								CBA > A_5			
			214									OUT						IN										CBA > A_6			
			215										OUT				IN											CBA > A_7			
			216											OUT		IN												CBA > A_8			
	t_{PHL1}		217	IN	4.5 V	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		GND	GND		CAB > B_1		40	
			218																									CAB > B_2			
			219																									CAB > B_3			
			220																									CAB > B_4			
			221																									CAB > B_5			
			222																									CAB > B_6			
			223																									CAB > B_7			
			224																									CAB > B_8			
			225	GND	GND	GND	OUT	OUT																	4.5 V	IN		CBA > A_1			
			226						OUT											IN								CBA > A_2			
			227							OUT											IN							CBA > A_3			
			228								OUT								IN								CBA > A_4			
			229									OUT						IN										CBA > A_5			
			230										OUT				IN											CBA > A_6			
			231											OUT		IN												CBA > A_7			
			232																									CBA > A_8			
	t_{PLH2}		233			4.5 V	IN	IN	IN	IN	IN	IN	IN	IN		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		GND	GND		A_1>B_1		23	
			234																									A_2>B_2			
			235																									A_3>B_3			
			236																									A_4>B_4			
			237																									A_5>B_5			
			238																									A_6>B_6			
			239																									A_7>B_7			
			240																									A_8>B_8			
			241			GND	OUT	OUT																				B_1>A_1			
			242						OUT											IN								B_2>A_2			
			243							OUT											IN							B_3>A_3			
			244								OUT								IN								B_4>A_4			
			245									OUT						IN										B_5>A_5			
			246										OUT				IN											B_6>A_6			
			247											OUT		IN												B_7>A_7			
			248																									B_8>A_8			
	t_{PHL2}		249			4.5 V	IN	IN	IN	IN	IN	IN	IN	IN		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT					A_1>B_1		25	
			250																									A_2>B_2			
			251																									A_3>B_3			
			252																									A_4>B_4			
			253																									A_5>B_5			
			254																									A_6>B_6			
			255																									A_7>B_7			
			256																									A_8>B_8			
			257			GND	OUT	OUT																				B_1>A_1			
			258						OUT											IN								B_2>A_2			
			259							OUT											IN							B_3>A_3			
			260								OUT								IN								B_4>A_4			
			261									OUT						IN										B_5>A_5			
			262										OUT				IN											B_6>A_6			
			263											OUT		IN												B_7>A_7			
			264																									B_8>A_8			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PLH3}	3003 (fig. 4)	265	4/	IN	4.5 V	4/	4/	4/	4/	4/	4/	4/	4/	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	5.0 V	SAB ▸ B_1	2	45	ns
			266																									SAB ▸ B_2			
			267																									SAB ▸ B_3			
			268																									SAB ▸ B_4			
			269																									SAB ▸ B_5			
			270																									SAB ▸ B_6			
			271																									SAB ▸ B_7			
			272																									SAB ▸ B_8			
			273	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT									4/	IN	4/		SBA ▸ A_1			
			274																									SBA ▸ A_2			
			275																									SBA ▸ A_3			
			276																									SBA ▸ A_4			
			277																									SBA ▸ A_5			
			278																									SBA ▸ A_6			
			279																									SBA ▸ A_7			
			280													4/	4/	4/	4/	4/	4/	4/						SBA ▸ A_8			
	t_{PHL3}		281	4/	IN	4.5 V	4/	4/	4/	4/	4/	4/	4/	4/										GND	GND			SAB ▸ B_1		40	
			282																									SAB ▸ B_2			
			283																									SAB ▸ B_3			
			284																									SAB ▸ B_4			
			285																									SAB ▸ B_5			
			286																									SAB ▸ B_6			
			287																									SAB ▸ B_7			
			288																									SAB ▸ B_8			
			289	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT									4/	IN	4/		SBA ▸ A_1			
			290																									SBA ▸ A_2			
			291																									SBA ▸ A_3			
			292																									SBA ▸ A_4			
			293																									SBA ▸ A_5			
			294																									SBA ▸ A_6			
			295																									SBA ▸ A_7			
			296													4/	4/	4/	4/	4/	4/	4/						SBA ▸ A_8			
	t_{PLH4}		297	5/	IN	4.5 V	5/	5/	5/	5/	5/	5/	5/	5/										GND	GND			SAB ▸ B_1		55	
			298																									SAB ▸ B_2			
			299																									SAB ▸ B_3			
			300																									SAB ▸ B_4			
			301																									SAB ▸ B_5			
			302																									SAB ▸ B_6			
			303																									SAB ▸ B_7			
			304																									SAB ▸ B_8			
			305	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT									5/	IN	5/		SBA ▸ A_1			
			306																									SBA ▸ A_2			
			307																									SBA ▸ A_3			
			308																									SBA ▸ A_4			
			309																									SBA ▸ A_5			
			310																									SBA ▸ A_6			
			311																									SBA ▸ A_7			
			312													5/	5/	5/	5/	5/	5/	5/						SBA ▸ A_8			
	t_{PHL4}		313	5/	IN	4.5 V	5/	5/	5/	5/	5/	5/	5/	5/										GND	GND			SAB ▸ B_1		30	
			314																									SAB ▸ B_2			
			315																									SAB ▸ B_3			
			316																									SAB ▸ B_4			
			317																									SAB ▸ B_5			
			318																									SAB ▸ B_6			
			319																									SAB ▸ B_7			
			320																									SAB ▸ B_8			
			321	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT									5/	IN	5/		SBA ▸ A_1			
			322																									SBA ▸ A_2			
			323																									SBA ▸ A_3			
			324																									SBA ▸ A_4			
			325																									SBA ▸ A_5			
			326																									SBA ▸ A_6			
			327																									SBA ▸ A_7			
			328													5/	5/	5/	5/	5/	5/	5/						SBA ▸ A_8			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}				
9 T_C = +25°C	t_{PZH2}	3003 (fig. 4)	329		GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	IN	GND		5.0 V	$\overline{G}$ to A_1	2	60	ns
			330																									$\overline{G}$ to A_2			
			331																									$\overline{G}$ to A_3			
			332																									$\overline{G}$ to A_4			
			333																									$\overline{G}$ to A_5			
			334																									$\overline{G}$ to A_6			
			335																									$\overline{G}$ to A_7			
			336																									$\overline{G}$ to A_8			
			337																									$\overline{G}$ to B_1			
			338																									$\overline{G}$ to B_2			
			339																									$\overline{G}$ to B_3			
			340																									$\overline{G}$ to B_4			
			341																									$\overline{G}$ to B_5			
			342																									$\overline{G}$ to B_6			
			343																									$\overline{G}$ to B_7			
			344																									$\overline{G}$ to B_8			
	t_{PZL2}		345			GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		GND	GND	GND	GND	GND	GND	GND	GND					$\overline{G}$ to A_1		70	
			346																									$\overline{G}$ to A_2			
			347																									$\overline{G}$ to A_3			
			348																									$\overline{G}$ to A_4			
			349																									$\overline{G}$ to A_5			
			350																									$\overline{G}$ to A_6			
			351																									$\overline{G}$ to A_7			
			352																									$\overline{G}$ to A_8			
			353																									$\overline{G}$ to B_1			
			354																									$\overline{G}$ to B_2			
			355																									$\overline{G}$ to B_3			
			356																									$\overline{G}$ to B_4			
			357																									$\overline{G}$ to B_5			
			358																									$\overline{G}$ to B_6			
			359																									$\overline{G}$ to B_7			
			360																									$\overline{G}$ to B_8			
	t_{PZH3}		361			IN	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND				DIR to B_1		90	
			362																									DIR to B_2			
			363																									DIR to B_3			
			364																									DIR to B_4			
			365																									DIR to B_5			
			366																									DIR to B_6			
			367																									DIR to B_7			
			368																									DIR to B_8			
			369																									DIR to A_1			
			370																									DIR to A_2			
			371																									DIR to A_3			
			372																									DIR to A_4			
			373																									DIR to A_5			
			374																									DIR to A_6			
			375																									DIR to A_7			
			376																									DIR to A_8			
	t_{PZL3}		377				GND	GND	GND	GND	GND	GND	GND	GND		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT					DIR to B_1		65	
			378																									DIR to B_2			
			379																									DIR to B_3			
			380																									DIR to B_4			
			381																									DIR to B_5			
			382																									DIR to B_6			
			383																									DIR to B_7			
			384																									DIR to B_8			
			385																									DIR to A_1			
			386																									DIR to A_2			
			387																									DIR to A_3			
			388																									DIR to A_4			
			389																									DIR to A_5			
			390																									DIR to A_6			
			391																									DIR to A_7			
			392																									DIR to A_8			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}				
9 T_C = +25°C	t_{PHZ2}	3003 (fig. 4)	393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408		GND " " " " " " " " " " " " " " "	4.5 V " " " " " " " GND " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	GND " " " " " " " " " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	IN " " " " " " " " " " " " " " "	GND " " " " " " " " " " " " " " "			5.0 V " " " " " " " " " " " " " " "	G ▸ B_1 G ▸ B_2 G ▸ B_3 G ▸ B_4 G ▸ B_5 G ▸ B_6 G ▸ B_7 G ▸ B_8 G ▸ A_1 G ▸ A_2 G ▸ A_3 G ▸ A_4 G ▸ A_5 G ▸ A_6 G ▸ A_7 G ▸ A_8	2 " " " " " " " " " " " " " " "	40 " " " " " " " " " " " " " " "	ns " " " " " " " " " " " " " " "
	t_{PLZ2}	"	409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424		" " " " " " " " " " " " " " " "	4.5 V " " " " " " " GND " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	" " " " " " " " " " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	GND " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "			" " " " " " " " " " " " " " " "	G ▸ B_1 G ▸ B_2 G ▸ B_3 G ▸ B_4 G ▸ B_5 G ▸ B_6 G ▸ B_7 G ▸ B_8 G ▸ A_1 G ▸ A_2 G ▸ A_3 G ▸ A_4 G ▸ A_5 G ▸ A_6 G ▸ A_7 G ▸ A_8	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "
	t_{PHZ3}	"	425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440		" " " " " " " " " " " " " " " "	IN " " " " " " " " " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	4.5 V " " " " " " " OUT " " " " " " "	" " " " " " " " " " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	OUT " " " " " " " 4.5 V " " " " " " "	GND " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "			" " " " " " " " " " " " " " " "	DIR ▸ B_1 DIR ▸ B_2 DIR ▸ B_3 DIR ▸ B_4 DIR ▸ B_5 DIR ▸ B_6 DIR ▸ B_7 DIR ▸ B_8 DIR ▸ A_1 DIR ▸ A_2 DIR ▸ A_3 DIR ▸ A_4 DIR ▸ A_5 DIR ▸ A_6 DIR ▸ A_7 DIR ▸ A_8	" " " " " " " " " " " " " " " "	15 " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "
	t_{PLZ3}	"	441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456		" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	GND " " " " " " " OUT " " " " " " "	" " " " " " " " " " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	OUT " " " " " " " GND " " " " " " "	GND " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "			" " " " " " " " " " " " " " " "	DIR ▸ B_1 DIR ▸ B_2 DIR ▸ B_3 DIR ▸ B_4 DIR ▸ B_5 DIR ▸ B_6 DIR ▸ B_7 DIR ▸ B_8 DIR ▸ A_1 DIR ▸ A_2 DIR ▸ A_3 DIR ▸ A_4 DIR ▸ A_5 DIR ▸ A_6 DIR ▸ A_7 DIR ▸ A_8	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "	" " " " " " " " " " " " " " " "

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}		Min	Max	
10 T_C = +125°C	t_{PLH1}																												2	39	ns
	t_{PHL1}																												"	52	"
	t_{PLH2}																												"	30	"
	t_{PHL2}																												"	33	"
	t_{PLH3}																												"	59	"
	t_{PHL3}																												"	52	"
	t_{PLH4}																												"	72	"
	t_{PHL4}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																											"	39	"
	t_{PZH2}																												"	78	"
	t_{PZL2}																												"	91	"
	t_{PZH3}																												"	65	"
	t_{PZL3}																												"	85	"
	t_{PHZ2}																												"	52	"
	t_{PLZ2}																												"	52	"
	t_{PHZ3}																												"	46	"
	t_{PLZ3}																												"	46	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																														

1/ Tests shall be performed in sequence, attributes data only.
2/ H > 1.5 V; L < 1.5 V.
3/ A = 3.0 V minimum; B = 0.0 V or GND.
4/ Prior to test, bus registers are loaded high by placing 4.5 V on bus data and applying one clock pulse (2.5 V/5.5 V, 0 V); the bus is then placed at GND for the duration of the test.
5/ Prior to test, bus registers are loaded low by placing GND on bus data and applying one clock pulse (2.5 V/5.5 V, 0 V); the bus is then placed at 4.5 V for the duration of the test.

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	G	SEL BA	CLK BA	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	0.5 V	0.5 V	2.0 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	GND	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	0.5 V	0.5 V	0.5 V	4.5 V	B_1	2.4		V
			2	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_2	"		"
			3	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_3	"		"
			4	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_4	"		"
			5	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_5	"		"
			6	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_6	"		"
			7	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_7	"		"
			8	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_8	"		"
			9	"	"	0.5 V	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	"	"	A_1	"		"
			10	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_2	"		"
			11	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_3	"		"
			12	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_4	"		"
			13	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_5	"		"
			14	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_6	"		"
			15	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_7	"		"
			16	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_8	"		"
	V_{OH2}	3006	17	"	"	2.0 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	"	"	"	"	B_1	2.0		"
			18	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_2	"		"
			19	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_3	"		"
			20	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_4	"		"
			21	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_5	"		"
			22	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_6	"		"
			23	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_7	"		"
			24	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_8	"		"
			25	"	"	0.5 V	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	"	"	A_1	"		"
			26	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_2	"		"
			27	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_3	"		"
			28	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_4	"		"
			29	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_5	"		"
			30	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_6	"		"
			31	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_7	"		"
			32	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_8	"		"
	V_{OL}	"	33	"	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	"	"	"	"	B_1		0.4	"
			34	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_2	"	"	"
			35	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_3	"	"	"
			36	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_4	"	"	"
			37	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_5	"	"	"
			38	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_6	"	"	"
			39	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_7	"	"	"
			40	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B_8	"	"	"
			41	"	"	0.5 V	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	A_1	"	"	"
			42	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_2	"	"	"
			43	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_3	"	"	"
			44	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_4	"	"	"
			45	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_5	"	"	"
			46	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_6	"	"	"
			47	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_7	"	"	"
			48	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A_8	"	"	"
	V_{IC}		49	-18 mA																								CAB		-1.5	"
			50	"	-18 mA																							SAB		"	"
			51	"	"	-18 mA																						DIR		"	"
			52	"	"	"	-18 mA																					A_1		"	"
			53	"	"	"	"	-18 mA																				A_2		"	"
			54	"	"	"	"	"	-18 mA																			A_3		"	"
			55	"	"	"	"	"	"	-18 mA																		A_4		"	"
			56	"	"	"	"	"	"	"	-18 mA																	A_5		"	"
			57	"	"	"	"	"	"	"	"	-18 mA																A_6		"	"
			58	"	"	"	"	"	"	"	"	"	-18 mA															A_7		"	"
			59	"	"	"	"	"	"	"	"	"	"	-18 mA														A_8		"	"
			60	"	"	"	"	"	"	"	"	"	"	"		-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA				B_8		"	"
			61	"	"	"	"	"	"	"	"	"	"	"		"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA				B_7		"	"
			62	"	"	"	"	"	"	"	"	"	"	"		"	"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA				B_6		"	"
			63	"	"	"	"	"	"	"	"	"	"	"		"	"	"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA				B_5		"	"
			64	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA				B_4		"	"
			65	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	-18 mA	-18 mA	-18 mA	-18 mA				B_3		"	"
			66	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	-18 mA	-18 mA	-18 mA				B_2		"	"
			67	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"	-18 mA	-18 mA				B_1		"	"
			68	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"	"	-18 mA				G		"	"
			69	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"	"	"				SBA		"	"
			70	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	"	"	"	"	"				CBA		"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	G	SEL BA	CLK BA	V_{CC}				
1 T_C = +25°C	I_{IL}	3009	71	0.4 V																						5.5 V	CAB	0	200	µA
		"	72		0.4 V																						"	SAB			"
		"	73			0.4 V																					"	DIR			"
		"	74				0.4 V																				"	A_1			"
		"	75					0.4 V																			"	A_2			"
		"	76						0.4 V																		"	A_3			"
		"	77							0.4 V																	"	A_4			"
		"	78								0.4 V																"	A_5			"
		"	79									0.4 V															"	A_6			"
		"	80										0.4 V														"	A_7			"
		"	81											0.4 V													"	A_8			"
		"	82													0.4 V											"	B_8			"
		"	83														0.4 V										"	B_7			"
		"	84															0.4 V									"	B_6			"
		"	85																0.4 V								"	B_5			"
		"	86																	0.4 V							"	B_4			"
		"	87																		0.4 V						"	B_3			"
		"	88																			0.4 V					"	B_2			"
		"	89																				0.4 V				"	B_1			"
		"	90																					0.4 V			"	G			"
		"	91																						0.4 Vo		"	SBA			"
		"	92																							0.4 V	"	CBA			"
	I_{IH1}	3010	93	2.7 V																							"	CAB		20	"
		"	94		2.7 V																						"	SAB			"
		"	95			2.7 V																					"	DIR			"
		"	96				2.7 V																				"	A_1			"
		"	97					2.7 V																			"	A_2			"
		"	98						2.7 V																		"	A_3			"
		"	99							2.7 V																	"	A_4			"
		"	100								2.7 V																"	A_5			"
		"	101									2.7 V															"	A_6			"
		"	102										2.7 V														"	A_7			"
		"	103											2.7 V													"	A_8			"
		"	104													2.7 V											"	B_8			"
		"	105														2.7 V										"	B_7			"
		"	106															2.7 V									"	B_6			"
		"	107																2.7 V								"	B_5			"
		"	108																	2.7 V							"	B_4			"
		"	109																		2.7 V						"	B_3			"
		"	110																			2.7 V					"	B_2			"
		"	111																				2.7 V				"	B_1			"
		"	112																					2.7 V			"	G			"
		"	113																						2.7 V		"	SBA			"
		"	114																							2.7 V	"	CBA			"
	I_{IH2}	"	115	5.5 V																							"	CAB		100	"
		"	116		5.5 V																						"	SAB			"
		"	117			5.5 V																					"	DIR			"
		"	118				5.5 V																				"	A_1			"
		"	119					5.5 V																			"	A_2			"
		"	120						5.5 V																		"	A_3			"
		"	121							5.5 V																	"	A_4			"
		"	122								5.5 V																"	A_5			"
		"	123									5.5 V															"	A_6			"
		"	124										5.5 V														"	A_7			"
		"	125											5.5 V													"	A_8			"
		"	126													5.5 V											"	B_8			"
		"	127														5.5 V										"	B_7			"
		"	128															5.5 V									"	B_6			"
		"	129																5.5 V								"	B_5			"
		"	130																	5.5 V							"	B_4			"
		"	131																		5.5 V						"	B_3			"
		"	132																			5.5 V					"	B_2			"
		"	133																				5.5 V				"	B_1			"
		"	134																					5.5 V			"	G			"
		"	135																						5.5 V		"	SBA			"
		"	136																							5.5 V	"	CBA			"

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	CLK AB	SEL AB	OIK	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}		Min	Max	
1 T_C = +25°C	I_{OZL}		137-152				0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	GND	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	2.0 V			5.5 V	A1-A8, B8-B1		20	µA
	I_{OZH}		153-168				2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V							400	
	I_{OS}	3011	169-184	GND	GND	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			-90	-225	mA
	I_{CCH}	3005	185			4.5 V	GND	GND	GND	GND	GND	GND	GND	GND														V_{CC}		145	
	I_{CCL}		186				4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V														V_{CC}		165	
	I_{CCZ}		187				GND	GND	GND	GND	GND	GND	GND	GND										4.5 V				V_{CC}		165	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																														
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																														
7 T_C = +25°C	Truth table tests 1/		188-200	B A	B A	A B A B	B A H L	B A H L	B A H L	B A H L	B A H L	B A H L	B A H L	B A H L	GND	H L B A	H L B A	H L B A	H L B A	H L B A	H L B A	H L B A	H L B A	B A	B A	B A B	4.5 V	2/ 3/			

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23[1]	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	G	SEL BA	CLK BA	V_{CC}				
8	Same tests and terminal conditions except T_C = +125°C and -55°C.																														
9 T_C = +25°C	t_{PLH1}	3003 (fig. 4)	201	IN	4.5 V	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	5.0 V	CAB > B_1	2	10	ns
			202	"	"	"	"								"									"				CAB > B_2	"	"	
			203	"	"	"	"								"									"				CAB > B_3	"	"	
			204	"	"	"	"								"									"				CAB > B_4	"	"	
			205	"	"	"	"								"									"				CAB > B_5	"	"	
			206	"	"	"	"								"									"				CAB > B_6	"	"	
			207	"	"	"	"								"									"				CAB > B_7	"	"	
			208	"	"	"	"								"									"				CAB > B_8	"	"	
			209	GND	GND	GND	OUT	OUT								OUT							IN	4.5 V	IN			CBA > A_1	"	"	
			210	"	"	"	"								"									"				CBA > A_2	"	"	
			211	"	"	"	"		OUT						"						IN			"				CBA > A_3	"	"	
			212	"	"	"	"			OUT					"					IN				"				CBA > A_4	"	"	
			213	"	"	"	"				OUT				"				IN					"				CBA > A_5	"	"	
			214	"	"	"	"					OUT			"									"				CBA > A_6	"	"	
			215	"	"	"	"						OUT		"									"				CBA > A_7	"	"	
			216	"	"	"	"							OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"				CBA > A_8	"	"	
	t_{PHL1}	"	217	IN	4.5 V	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	"	CAB > B_1	"	45	"
			218	"	"	"	"								"									"				CAB > B_2	"	"	
			219	"	"	"	"								"									"				CAB > B_3	"	"	
			220	"	"	"	"								"									"				CAB > B_4	"	"	
			221	"	"	"	"								"									"				CAB > B_5	"	"	
			222	"	"	"	"								"									"				CAB > B_6	"	"	
			223	"	"	"	"								"									"				CAB > B_7	"	"	
			224	"	"	"	"								"									"				CAB > B_8	"	"	
			225	GND	GND	GND	OUT	OUT								OUT							IN	4.5 V	IN			CBA > A_1	"	"	
			226	"	"	"	"								"									"				CBA > A_2	"	"	
			227	"	"	"	"		OUT						"						IN			"				CBA > A_3	"	"	
			228	"	"	"	"			OUT					"					IN				"				CBA > A_4	"	"	
			229	"	"	"	"				OUT				"				IN					"				CBA > A_5	"	"	
			230	"	"	"	"					OUT			"									"				CBA > A_6	"	"	
			231	"	"	"	"						OUT		"									"				CBA > A_7	"	"	
			232	"	"	"	"							OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"				CBA > A_8	"	"	
	t_{PLH2}	"	233	"	"	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	"	A_1>B_1	"	23	"
			234	"	"	"	"								"									"				A_2>B_2	"	"	
			235	"	"	"	"								"									"				A_3>B_3	"	"	
			236	"	"	"	"								"									"				A_4>B_4	"	"	
			237	"	"	"	"								"									"				A_5>B_5	"	"	
			238	"	"	"	"								"									"				A_6>B_6	"	"	
			239	"	"	"	"								"									"				A_7>B_7	"	"	
			240	"	"	"	"								"									"				A_8>B_8	"	"	
			241	"	"	GND	OUT	OUT								OUT							IN	"	IN			B_1>A_1	"	"	
			242	"	"	"	"								"									"				B_2>A_2	"	"	
			243	"	"	"	"		OUT						"						IN			"				B_3>A_3	"	"	
			244	"	"	"	"			OUT					"					IN				"				B_4>A_4	"	"	
			245	"	"	"	"				OUT				"				IN					"				B_5>A_5	"	"	
			246	"	"	"	"					OUT			"									"				B_6>A_6	"	"	
			247	"	"	"	"						OUT		"									"				B_7>A_7	"	"	
			248	"	"	"	"							OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"				B_8>A_8	"	"	
	t_{PHL2}	"	249	"	"	4.5 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	"	"	A_1>B_1	"	30	"
			250	"	"	"	"								"									"				A_2>B_2	"	"	
			251	"	"	"	"								"									"				A_3>B_3	"	"	
			252	"	"	"	"								"									"				A_4>B_4	"	"	
			253	"	"	"	"								"									"				A_5>B_5	"	"	
			254	"	"	"	"								"									"				A_6>B_6	"	"	
			255	"	"	"	"								"									"				A_7>B_7	"	"	
			256	"	"	"	"								"									"				A_8>B_8	"	"	
			257	"	"	GND	OUT	OUT								OUT							IN	"	IN			B_1>A_1	"	"	
			258	"	"	"	"								"									"				B_2>A_2	"	"	
			259	"	"	"	"		OUT						"						IN			"				B_3>A_3	"	"	
			260	"	"	"	"			OUT					"					IN				"				B_4>A_4	"	"	
			261	"	"	"	"				OUT				"				IN					"				B_5>A_5	"	"	
			262	"	"	"	"					OUT			"									"				B_6>A_6	"	"	
			263	"	"	"	"						OUT		"									"				B_7>A_7	"	"	
			264	"	"	"	"							OUT	"	IN	IN	IN	IN	IN	IN	IN	IN	"				B_8>A_8	"	"	

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	G	SEL BA	CLK BA	V_{CC}				
9	t_{PLH3}	3003 (Fig. 4)	265	4/	IN	4.5 V	4/	4/	4/	4/	4/	4/	4/	4/	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	5.0 V	$SAB \rightarrow B_1$	2	60	ns
T_C = +25°C			266																									$SAB \rightarrow B_2$			
			267																									$SAB \rightarrow B_3$			
			268																									$SAB \rightarrow B_4$			
			269																									$SAB \rightarrow B_5$			
			270																									$SAB \rightarrow B_6$			
			271																									$SAB \rightarrow B_7$			
			272																									$SAB \rightarrow B_8$			
			273	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	OUT	OUT	OUT	4/		IN	4/		$SBA \rightarrow A_1$			
			274																									$SBA \rightarrow A_2$			
			275																									$SBA \rightarrow A_3$			
			276																									$SBA \rightarrow A_4$			
			277																									$SBA \rightarrow A_5$			
			278																									$SBA \rightarrow A_6$			
			279																									$SBA \rightarrow A_7$			
			280																									$SBA \rightarrow A_8$			
	t_{PHL3}		281	4/	IN	4.5 V	4/	4/	4/	4/	4/	4/	4/	4/		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND		$SAB \rightarrow B_1$		45	
			282																									$SAB \rightarrow B_2$			
			283																									$SAB \rightarrow B_3$			
			284																									$SAB \rightarrow B_4$			
			285																									$SAB \rightarrow B_5$			
			286																									$SAB \rightarrow B_6$			
			287																									$SAB \rightarrow B_7$			
			288																									$SAB \rightarrow B_8$			
			289	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	OUT	OUT	OUT	4/		IN	4/		$SBA \rightarrow A_1$			
			290																									$SBA \rightarrow A_2$			
			291																									$SBA \rightarrow A_3$			
			292																									$SBA \rightarrow A_4$			
			293																									$SBA \rightarrow A_5$			
			294																									$SBA \rightarrow A_6$			
			295																									$SBA \rightarrow A_7$			
			296																									$SBA \rightarrow A_8$			
	t_{PLH4}		297	5/	IN	4.5 V	5/	5/	5/	5/	5/	5/	5/	5/		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND		$SAB \rightarrow B_1$			
			298																									$SAB \rightarrow B_2$			
			299																									$SAB \rightarrow B_3$			
			300																									$SAB \rightarrow B_4$			
			301																									$SAB \rightarrow B_5$			
			302																									$SAB \rightarrow B_6$			
			303																									$SAB \rightarrow B_7$			
			304																									$SAB \rightarrow B_8$			
			305	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	OUT	OUT	OUT	5/		IN	5/		$SBA \rightarrow A_1$			
			306																									$SBA \rightarrow A_2$			
			307																									$SBA \rightarrow A_3$			
			308																									$SBA \rightarrow A_4$			
			309																									$SBA \rightarrow A_5$			
			310																									$SBA \rightarrow A_6$			
			311																									$SBA \rightarrow A_7$			
			312																									$SBA \rightarrow A_8$			
	t_{PHL4}		313	5/	IN	4.5 V	5/	5/	5/	5/	5/	5/	5/	5/		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND		$SAB \rightarrow B_1$			
			314																									$SAB \rightarrow B_2$			
			315																									$SAB \rightarrow B_3$			
			316																									$SAB \rightarrow B_4$			
			317																									$SAB \rightarrow B_5$			
			318																									$SAB \rightarrow B_6$			
			319																									$SAB \rightarrow B_7$			
			320																									$SAB \rightarrow B_8$			
			321	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	OUT	OUT	OUT	5/		IN	5/		$SBA \rightarrow A_1$			
			322																									$SBA \rightarrow A_2$			
			323																									$SBA \rightarrow A_3$			
			324																									$SBA \rightarrow A_4$			
			325																									$SBA \rightarrow A_5$			
			326																									$SBA \rightarrow A_6$			
			327																									$SBA \rightarrow A_7$			
			328																									$SBA \rightarrow A_8$			

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883, method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	G	SEL BA	CLK BA	V_{CC}				
9 T_C = +25°C	t_{PZH2}	3003 (fig. 4)	329		GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	GND	GND	GND	GND	GND	GND	IN	GND		5.0 V	G > A_1	2	55	ns
			330		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_2	"	"	"
			331		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_3	"	"	"
			332		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_4	"	"	"
			333		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_5	"	"	"
			334		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_6	"	"	"
			335		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_7	"	"	"
			336		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_8	"	"	"
			337		"	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"		"	G > B_1	"	"	"
			338		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_2	"	"	"
			339		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_3	"	"	"
			340		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_4	"	"	"
			341		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_5	"	"	"
			342		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_6	"	"	"
			343		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_7	"	"	"
			344		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_8	"	"	"
	t_{PZL2}	"	345		"	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"		"	G > A_1	"	60	"
			346		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_2	"	"	"
			347		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_3	"	"	"
			348		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_4	"	"	"
			349		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_5	"	"	"
			350		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_6	"	"	"
			351		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_7	"	"	"
			352		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > A_8	"	"	"
			353		"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"		"	G > B_1	"	"	"
			354		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_2	"	"	"
			355		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_3	"	"	"
			356		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_4	"	"	"
			357		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_5	"	"	"
			358		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_6	"	"	"
			359		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_7	"	"	"
			360		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	G > B_8	"	"	"
	t_{PZH3}	"	361		"	TN	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	"		"	DIR > B_1	"	45	"
			362		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_2	"	"	"
			363		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_3	"	"	"
			364		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_4	"	"	"
			365		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_5	"	"	"
			366		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_6	"	"	"
			367		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_7	"	"	"
			368		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_8	"	"	"
			369		"	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	"	"		"	DIR > A_1	"	"	"
			370		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_2	"	"	"
			371		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_3	"	"	"
			372		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_4	"	"	"
			373		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_5	"	"	"
			374		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_6	"	"	"
			375		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_7	"	"	"
			376		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_8	"	"	"
	t_{PZL3}	"	377		"	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	OUT	OUT	4.5 V	4.5 V	"	"		"	DIR > B_1	"	50	"
			378		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_2	"	"	"
			379		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_3	"	"	"
			380		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_4	"	"	"
			381		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_5	"	"	"
			382		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_6	"	"	"
			383		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_7	"	"	"
			384		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > B_8	"	"	"
			385		"	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"		"	DIR > A_1	"	"	"
			386		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_2	"	"	"
			387		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_3	"	"	"
			388		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_4	"	"	"
			389		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_5	"	"	"
			390		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_6	"	"	"
			391		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_7	"	"	"
			392		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"		"	DIR > A_8	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}				
9 T_C = +25°C	t_{PHZ2}	3003 (Fig. 4)	393		GND	4.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	IN	GND		5.0 V	$\overline{G}$ to B_1	2	50	ns
			394		"	"									"									"	"		"	$\overline{G}$ to B_2			
			395		"	"									"									"	"		"	$\overline{G}$ to B_3			
			396		"	"									"									"	"		"	$\overline{G}$ to B_4			
			397		"	"									"									"	"		"	$\overline{G}$ to B_5			
			398		"	"									"									"	"		"	$\overline{G}$ to B_6			
			399		"	"									"									"	"		"	$\overline{G}$ to B_7			
			400		"	"									"									"	"		"	$\overline{G}$ to B_8			
			401		"	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	GND	GND	GND	GND	GND	GND	GND	GND	"	"		"	$\overline{G}$ to A_1			
			402		"	"									"									"	"		"	$\overline{G}$ to A_2			
			403		"	"									"									"	"		"	$\overline{G}$ to A_3			
			404		"	"									"									"	"		"	$\overline{G}$ to A_4			
			405		"	"									"									"	"		"	$\overline{G}$ to A_5			
			406		"	"									"									"	"		"	$\overline{G}$ to A_6			
			407		"	"									"									"	"		"	$\overline{G}$ to A_7			
			408		"	"									"									"	"		"	$\overline{G}$ to A_8			
	t_{PLZ2}	"	409		"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"		"	$\overline{G}$ to B_1		40	"
			410		"	"									"									"	"		"	$\overline{G}$ to B_2			
			411		"	"									"									"	"		"	$\overline{G}$ to B_3			
			412		"	"									"									"	"		"	$\overline{G}$ to B_4			
			413		"	"									"									"	"		"	$\overline{G}$ to B_5			
			414		"	"									"									"	"		"	$\overline{G}$ to B_6			
			415		"	"									"									"	"		"	$\overline{G}$ to B_7			
			416		"	"									"									"	"		"	$\overline{G}$ to B_8			
			417		"	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"		"	$\overline{G}$ to A_1			
			418		"	"									"									"	"		"	$\overline{G}$ to A_2			
			419		"	"									"									"	"		"	$\overline{G}$ to A_3			
			420		"	"									"									"	"		"	$\overline{G}$ to A_4			
			421		"	"									"									"	"		"	$\overline{G}$ to A_5			
			422		"	"									"									"	"		"	$\overline{G}$ to A_6			
			423		"	"									"									"	"		"	$\overline{G}$ to A_7			
			424		"	"									"									"	"		"	$\overline{G}$ to A_8			
	t_{PHZ3}	"	425		"	IN	GND	GND	GND	GND	GND	GND	GND	GND	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	"		"	DIR to B_1		40	"
			426		"	"									"									"	"		"	DIR to B_2			
			427		"	"									"									"	"		"	DIR to B_3			
			428		"	"									"									"	"		"	DIR to B_4			
			429		"	"									"									"	"		"	DIR to B_5			
			430		"	"									"									"	"		"	DIR to B_6			
			431		"	"									"									"	"		"	DIR to B_7			
			432		"	"									"									"	"		"	DIR to B_8			
			433		"	"									"									"	"		"	DIR to A_1			
			434		"	"									"									"	"		"	DIR to A_2			
			435		"	"									"									"	"		"	DIR to A_3			
			436		"	"									"									"	"		"	DIR to A_4			
			437		"	"									"									"	"		"	DIR to A_5			
			438		"	"									"									"	"		"	DIR to A_6			
			439		"	"									"									"	"		"	DIR to A_7			
			440		"	"									"									"	"		"	DIR to A_8			
	t_{PLZ3}	"	441		"	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"		"	DIR to B_1		35	"
			442		"	"									"									"	"		"	DIR to B_2			
			443		"	"									"									"	"		"	DIR to B_3			
			444		"	"									"									"	"		"	DIR to B_4			
			445		"	"									"									"	"		"	DIR to B_5			
			446		"	"									"									"	"		"	DIR to B_6			
			447		"	"									"									"	"		"	DIR to B_7			
			448		"	"									"									"	"		"	DIR to B_8			
			449		"	"									"									"	"		"	DIR to A_1			
			450		"	"									"									"	"		"	DIR to A_2			
			451		"	"									"									"	"		"	DIR to A_3			
			452		"	"									"									"	"		"	DIR to A_4			
			453		"	"									"									"	"		"	DIR to A_5			
			454		"	"									"									"	"		"	DIR to A_6			
			455		"	"									"									"	"		"	DIR to A_7			
			456		"	"									"									"	"		"	DIR to A_8			

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high > 2.0 V; low < 0.5 V; or open)

Subgroup	Symbol	MIL-STD-883-method	Case L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	CLK AB	SEL AB	DIR	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	GND	B_8	B_7	B_6	B_5	B_4	B_3	B_2	B_1	$\overline{G}$	SEL BA	CLK BA	V_{CC}		Min	Max	
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																											2	39	ns
	t_{PHL1}																												"	59	"
	t_{PLH2}																												"	30	"
	t_{PHL2}																												"	39	"
	t_{PLH3}																												"	78	"
	t_{PHL3}																												"	59	"
	t_{PLH4}																												"	59	"
	t_{PHL4}																												"	59	"
	t_{PZH2}																												"	72	"
	t_{PZL2}																												"	78	"
	t_{PZH3}																												"	59	"
	t_{PZL3}																												"	65	"
	t_{PHZ2}																												"	65	"
	t_{PLZ2}																												"	52	"
	t_{PHZ3}																												"	52	"
	t_{PLZ3}																												"	46	"
11 T_C = -55°C 1/		Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																													

1/ Tests shall be performed in sequence, attributes data only.
2/ H > 1.5 V; L < 1.5 V.
3/ A = 3.0 V minimum; B = 0.0 V or GND.
4/ Prior to test, bus registers are loaded low by placing GND on bus data and applying one clock pulse (2.5 V/5.5 V / 0 V); the bus data is then placed at 4.5 V for the duration of the test.
5/ Prior to test, bus registers are loaded high by placing 4.5 V on bus data and applying one clock pulse (2.5 V/5.5 V / 0 V); the bus data is then placed at GND for the duration of the test.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Subgroups 3 and 4 shall be added to the group C inspection parameters for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
c. Steady-state life test (method 1005 of MIL-STD-883) conditions:
   (1) Test condition D, using the circuit shown on figure 3, or equivalent.
   (2) T_A = +125°C minimum.
   (3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.6 Inclusion with other detail specifications. For qualification and quality conformance inspection purposes, devices covered by this specification may be treated as though they were included on the same detail specification as devices covered by MIL-M-38510/324. In addition, if a manufacturer is already part I qualified for type 32402, and if the respective device on this specification (MIL-M-38510/328) is designed and manufactured identically (same die, same process, same screening) in all respects except electrical testing, then device type 32802 may be part I qualified by conducting only group A electrical tests, with the approval of the qualifying activity including subgroups A-10 and A-11, and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.
f. Requirements for product assurance options.
g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part numbers. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.
h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal
t_{PHZ}	Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high impedance (off) state.
t_{PLZ}	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high impedance (off) state to the defined high level.
t_{PZL}	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B, (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS242
02	54LS243
03	54LS245
04	54LS645
05	54LS648

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6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device types	Circuits				
	A	B	C	D	E
	Texas Instruments	Signetics Corp.	National Company	Raytheon Company	Motorola Inc.
01	X	X	X	X	X
02	X	X	X	X	X
03	X	X			X
04	X				
05	X				

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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